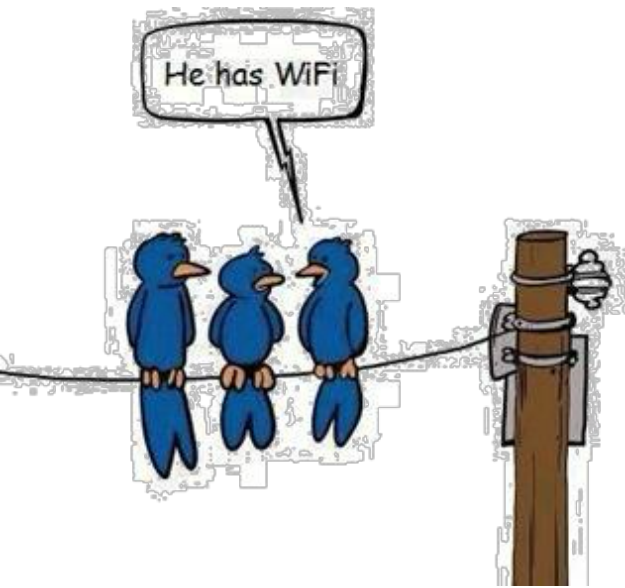


ECSE 4560/ 6560

Modern Communication Systems

Lecture 8: OFDM Introduction



Ish Jain, Fall 2025



Announcements

- Join Piazza: <https://piazza.com/rpi/fall2025/ecse45606560>
- Per-Class Quiz
 - Posted on LMS; Due Today 11:59 PM
- Attendance Link: <https://forms.office.com/r/p6P9YrN1d8>
 - Today's Magic Word: **OFDM**
- Projects
 - Project Topics posted on Piazza
 - Group formation [MCS Fall25 project.xlsx](#)
 - Proposal due extended **10/3**
 - 1 page: Project Title, Abstract, and Evaluation Plan
- HW2/Lab2 due Mon 9/29 (extended Wed 10/1)
 - Issues with PlutoSDR? Today's office hours

Today's Class: Multipath Channel (Cont.)+OFDM

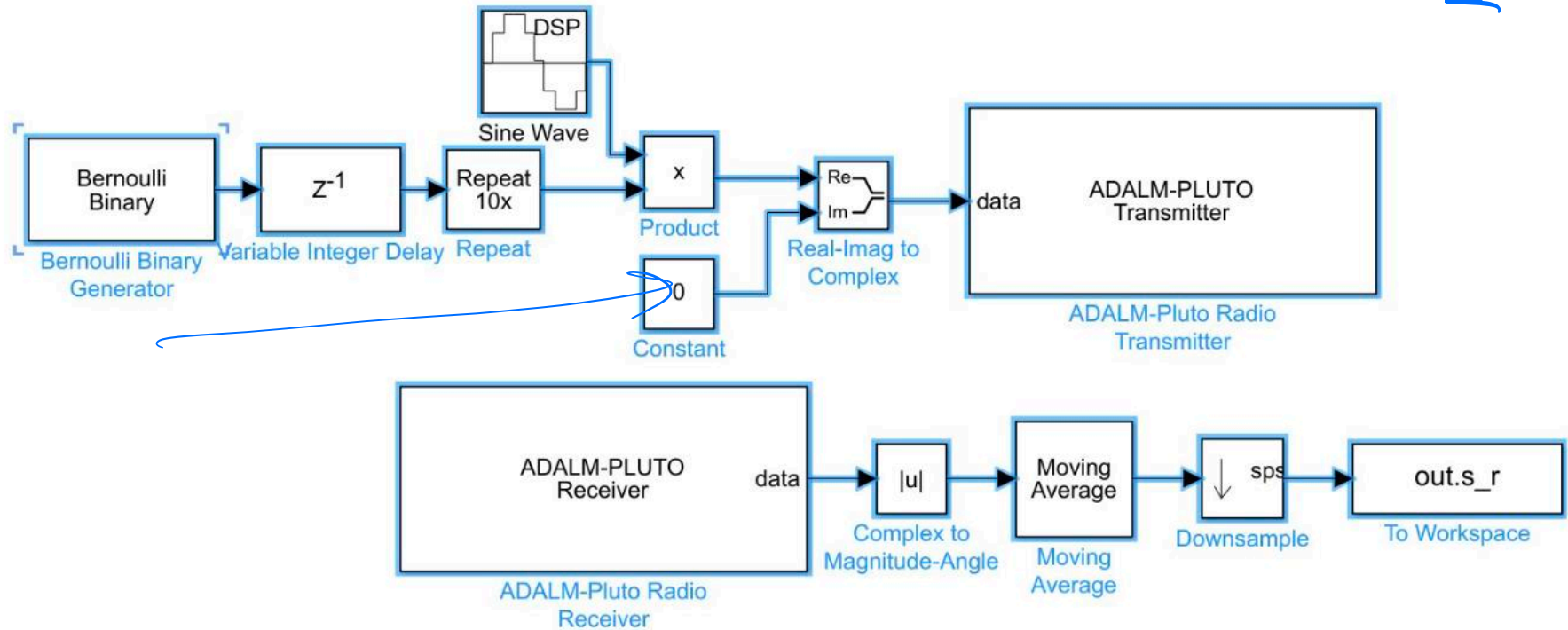
- Multipath Channel
 - Power-delay profile
 - Inter-symbol interference
 - Frequency selective fading
 - Narrowband and wideband channel
- OFDM

Hardware Pluto+ SDR



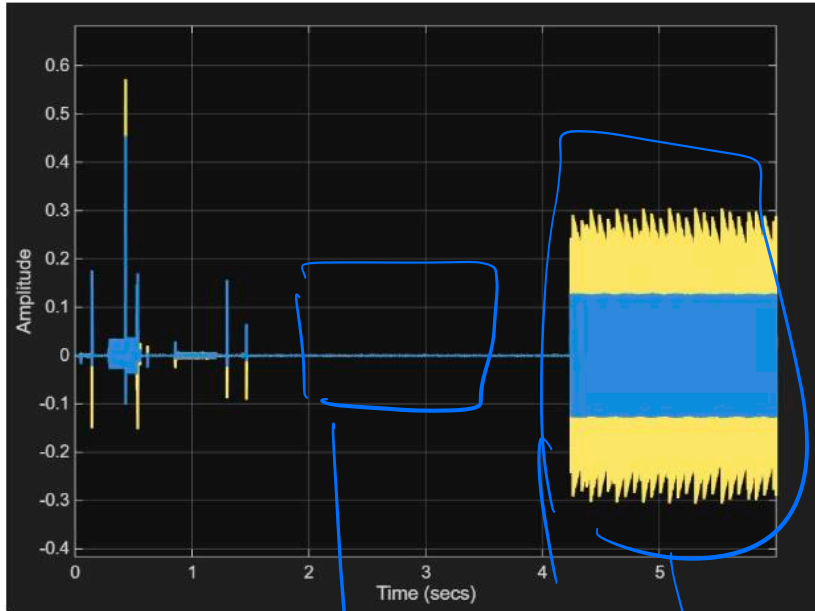
Hardware Pluto+ SDR

0015
0, 1



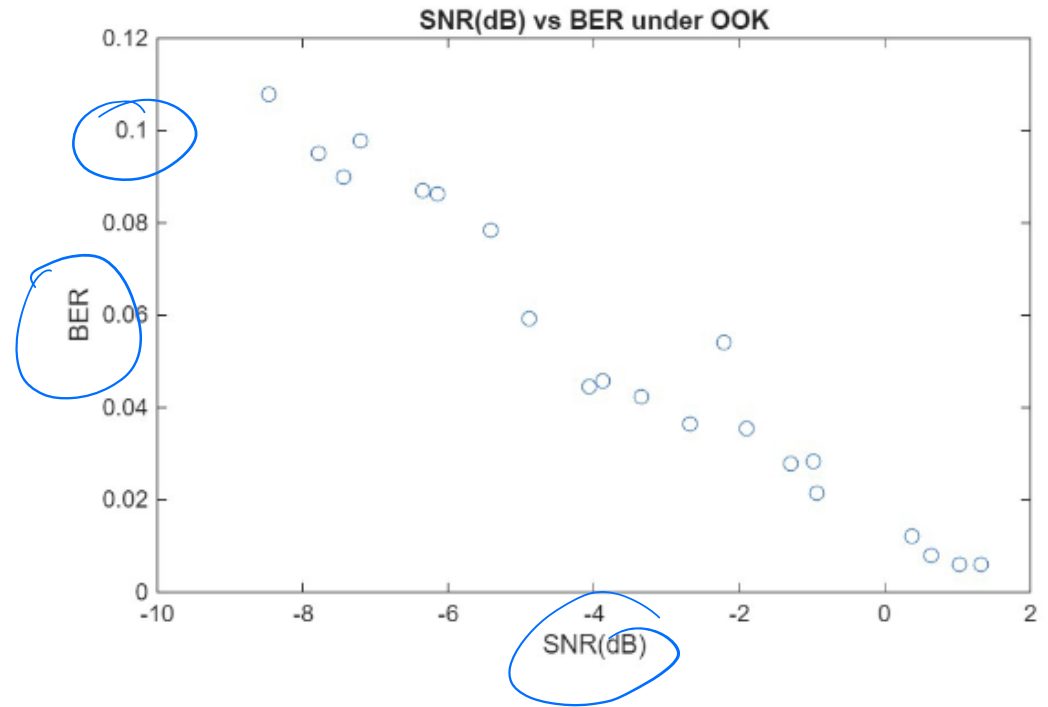
Hardware Pluto+ SDR

yes 0.1 BER
10% symbols are error



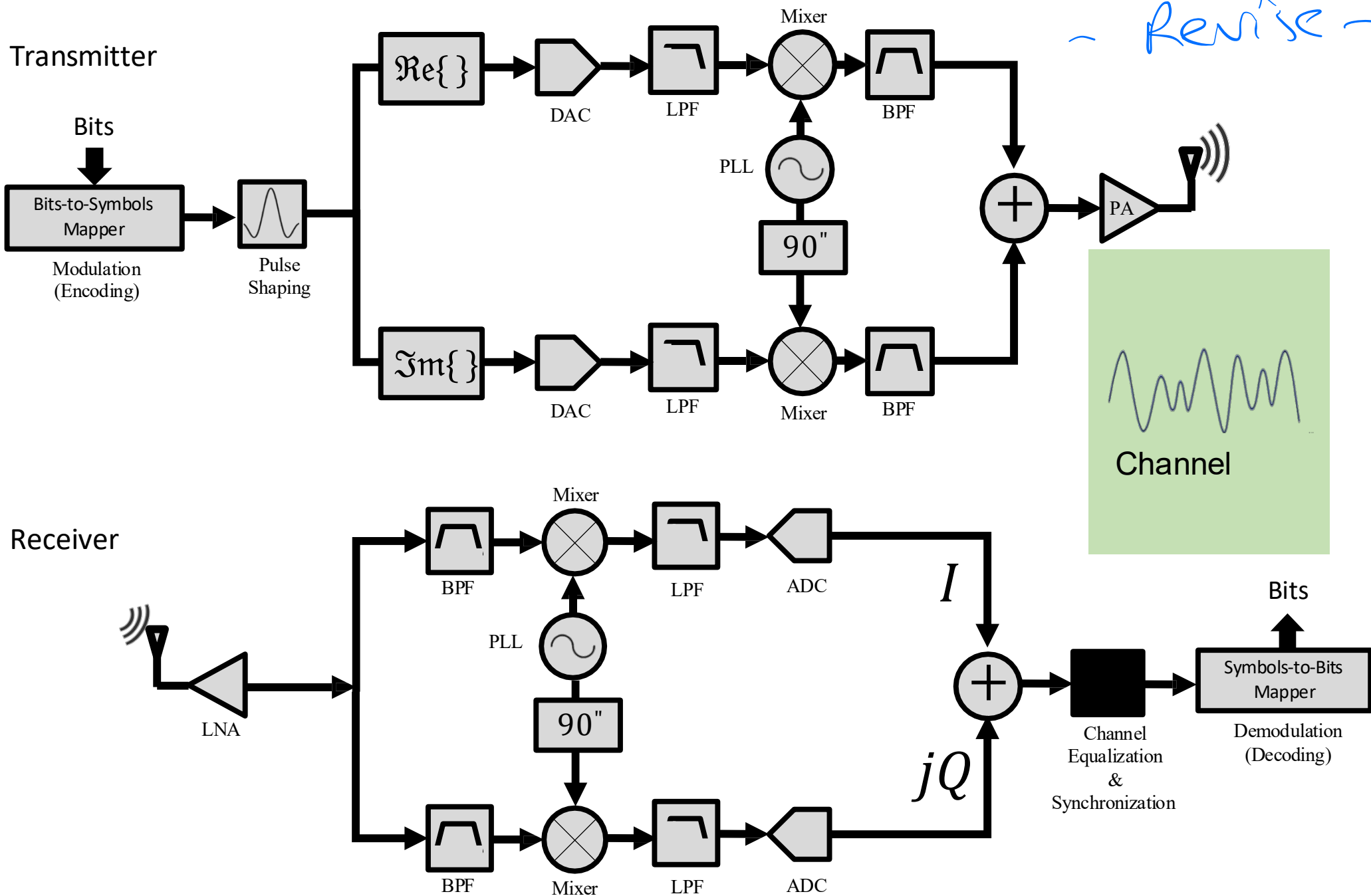
zero
noise

S/N
A

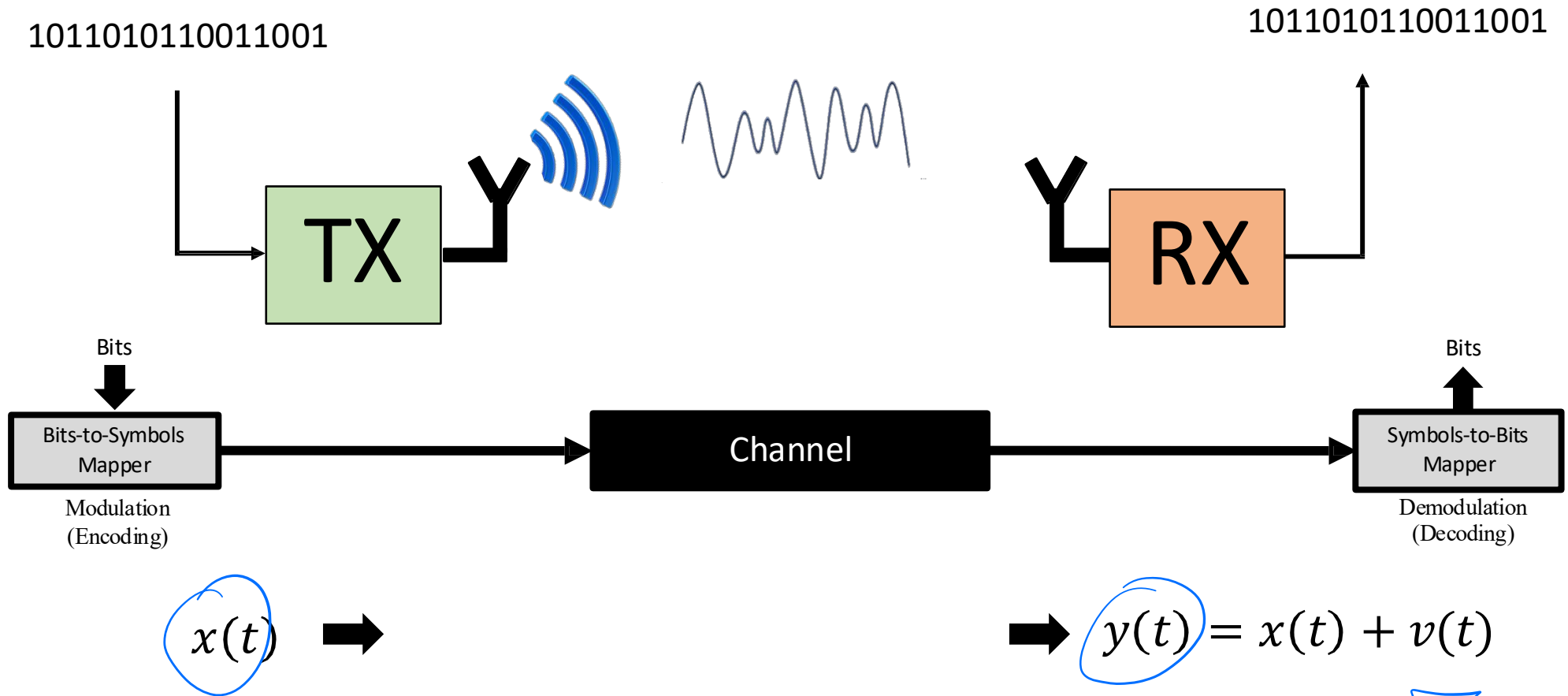


Signal power
noise power

Wireless Communication System



The Channel



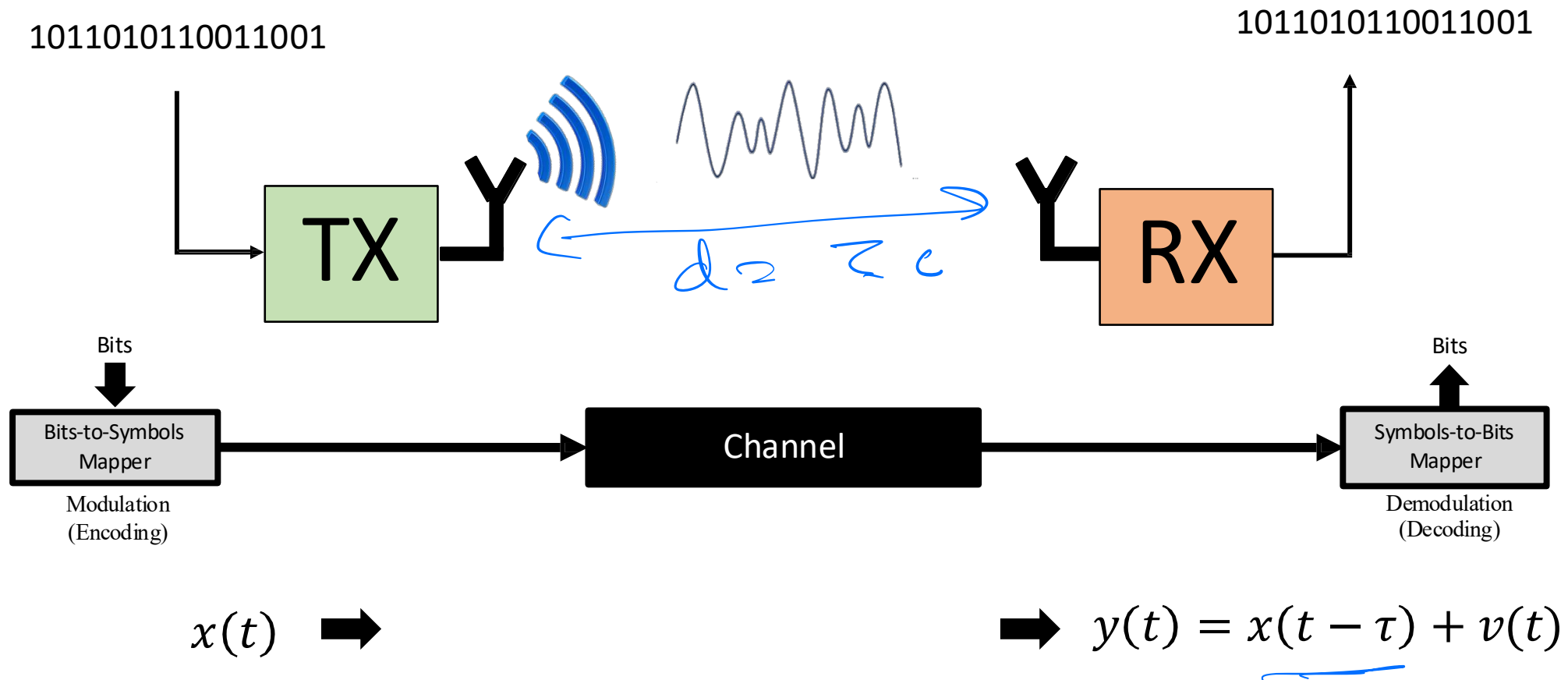
Channel adds noise (AWGN)!

$$v(t) \sim N(0, \sigma^2)$$

real.

$$\sigma^2 = \frac{N_0}{2}$$

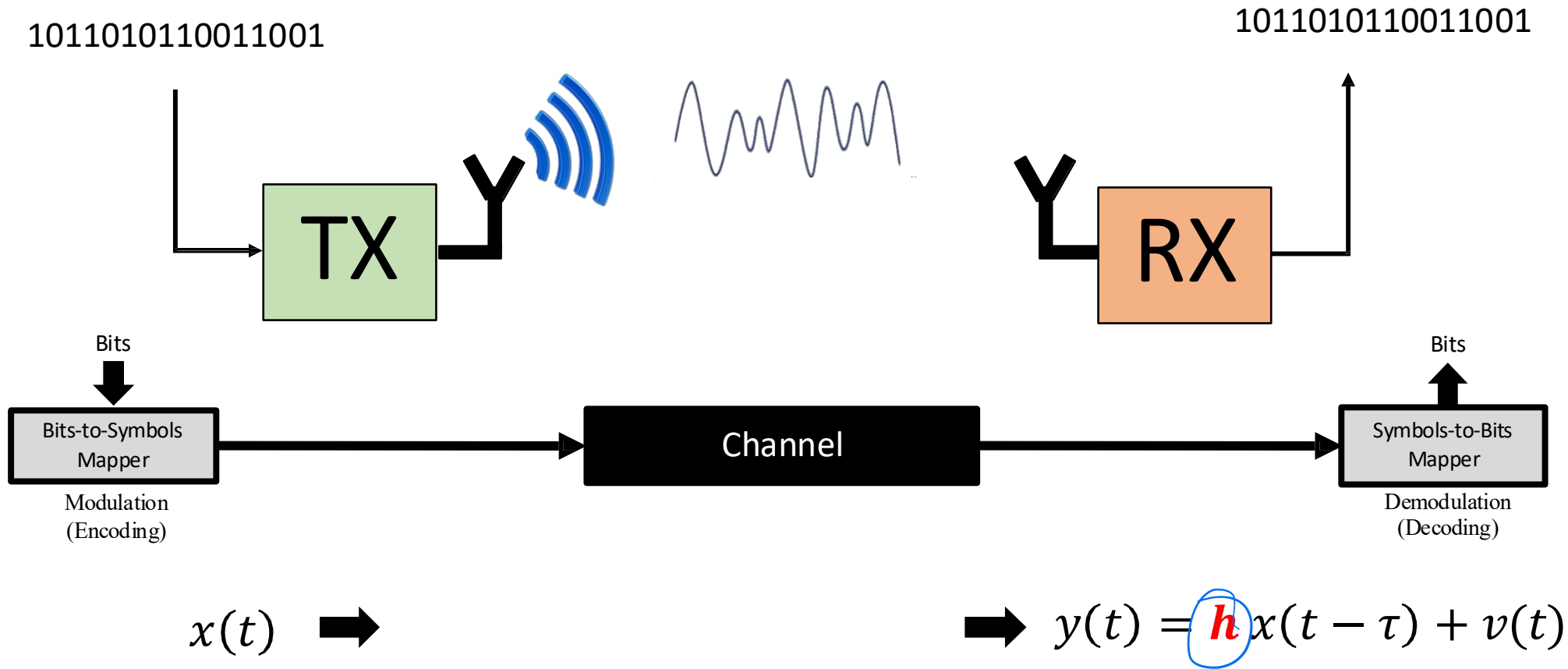
The Channel



Channel delays the signal!

$$\tau = \frac{d}{c}$$

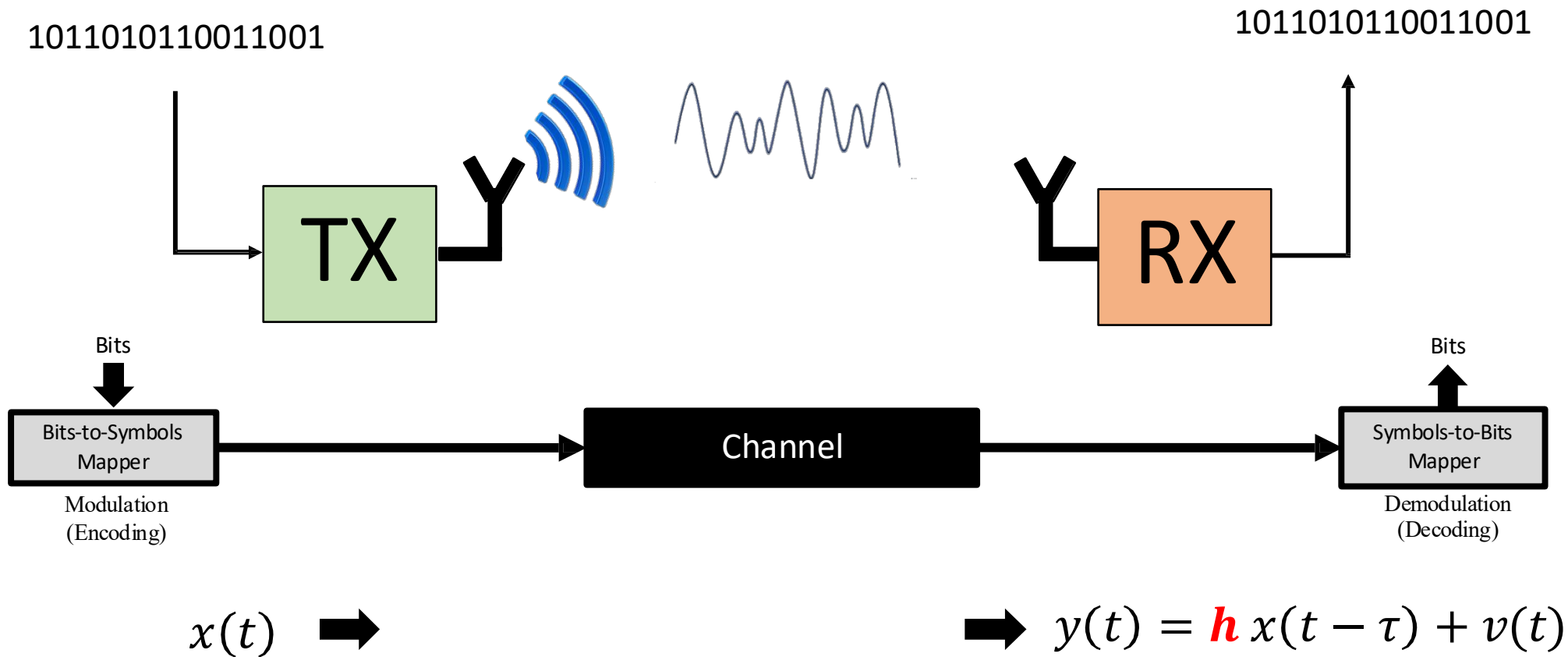
The Channel



Channel attenuates the signal (Pathloss)

$$P_{\text{RX}} = G_{\text{TX}} G_{\text{RX}} \frac{\lambda^2}{(4\pi d)^2} P_{\text{TX}} \Rightarrow |h| \propto \frac{\lambda}{d}$$

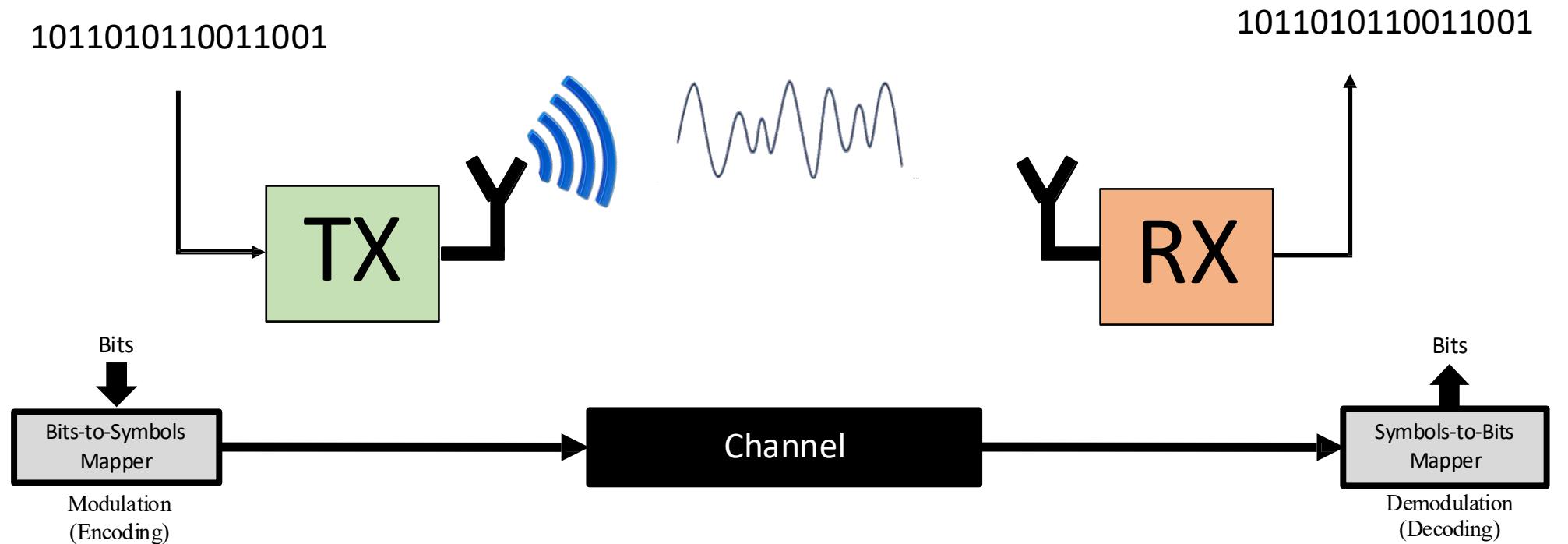
The Channel



Channel rotates the signal (Adds Phase)

$$h \propto \frac{\lambda}{d} e^{j\phi}$$

The Channel



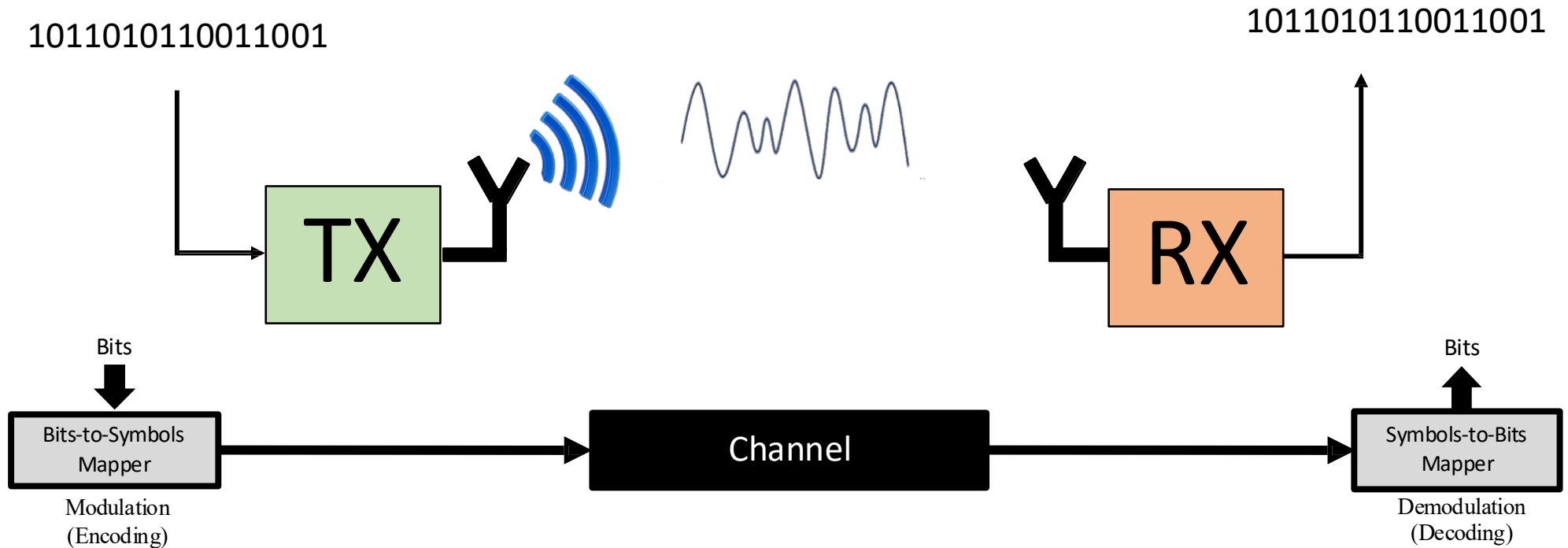
$$x(t) \Rightarrow$$

$$\Rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

$$x(t) \times e^{-j2\pi f_c t} \Rightarrow |h| x(t - \tau) e^{-j2\pi f_c (t - \tau)} \Rightarrow x(t - \tau) e^{j2\pi f_c \tau} \Rightarrow |h| x(t - \tau) e^{j2\pi f_c \tau}$$

$$h \propto \frac{\lambda}{d} e^{j\phi} \rightarrow \phi = 2\pi f_c \tau = 2\pi \frac{c}{\lambda} \frac{d}{c} = 2\pi \frac{d}{\lambda} \rightarrow h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel



$$x(t) \rightarrow$$

$$\rightarrow y(t) = \mathbf{h} x(t - \tau) + v(t)$$

Channel:

- Adds Noise
- Delays the Signal
- Attenuates the Signal
- Rotates the Phase of the Signal

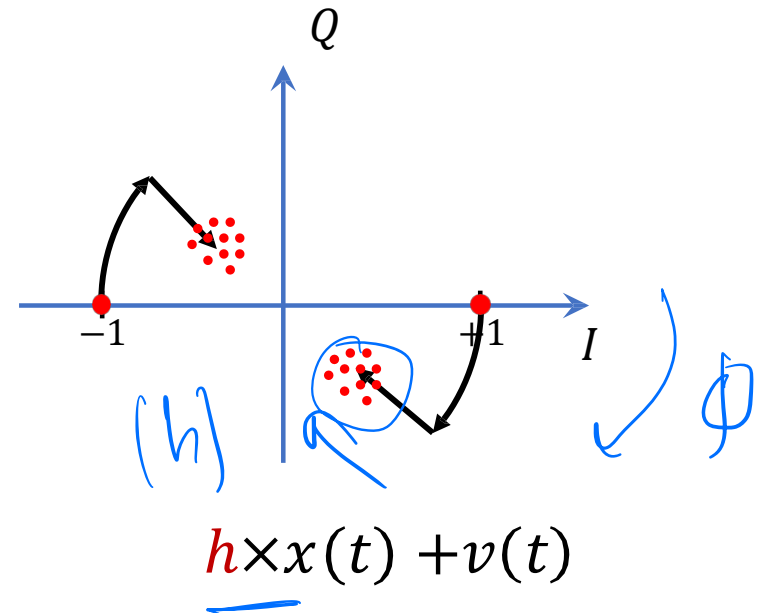
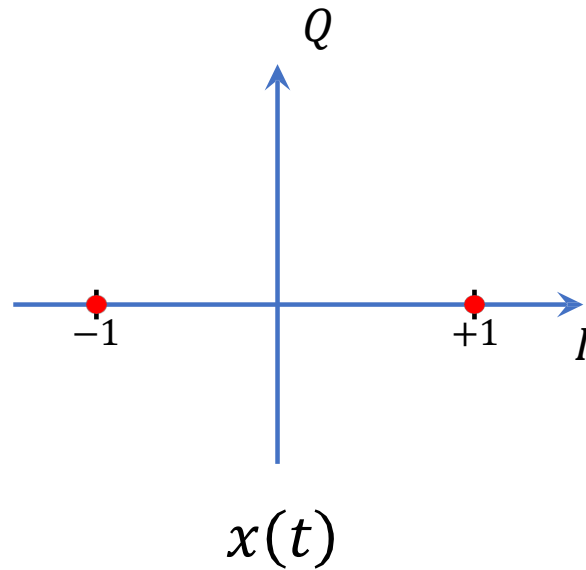
$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

The Channel

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



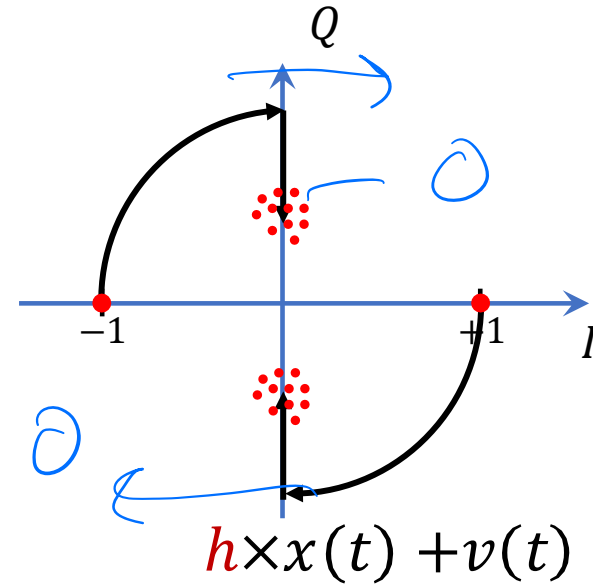
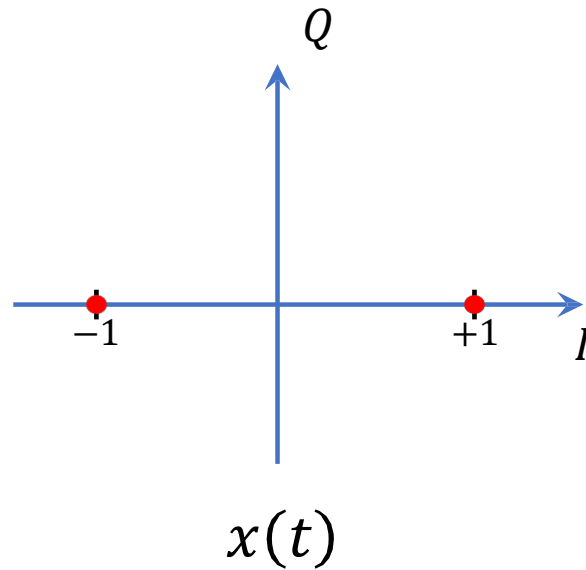
The Channel

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$

50%
BER

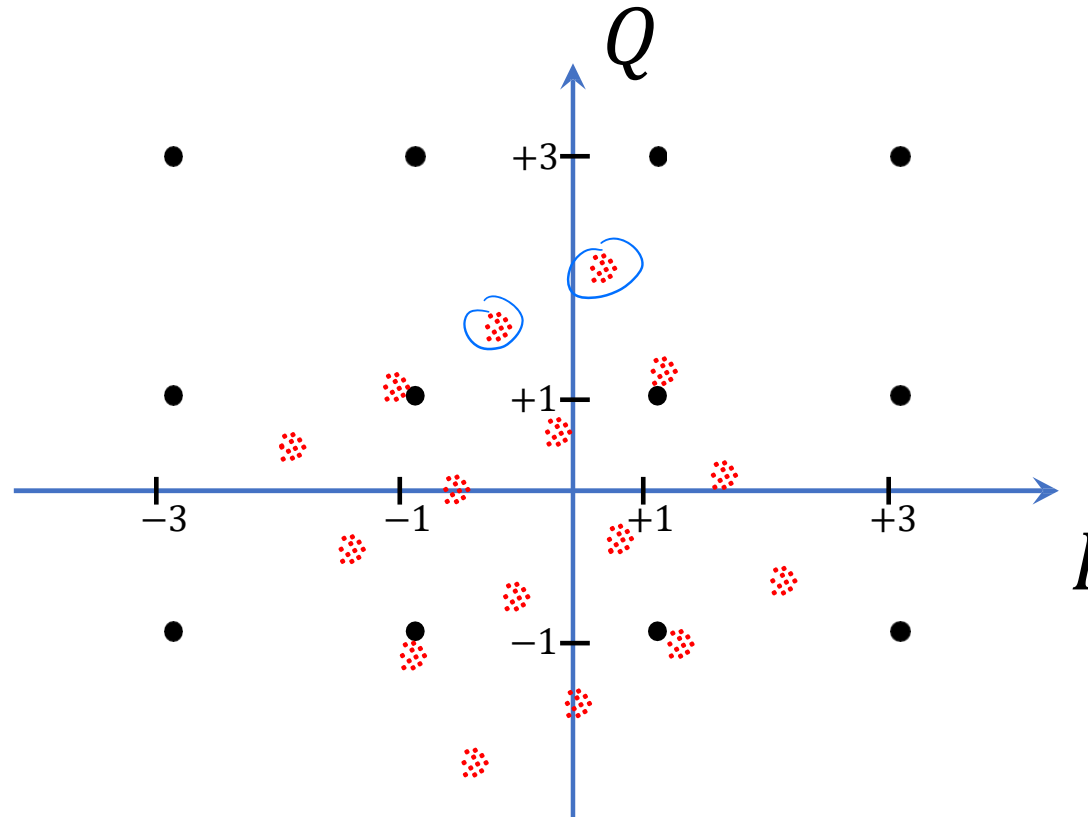


50% BER

The Channel

Consider QAM Modulation

16 QAM



Demodulating correctly requires COHERENCE!
i.e., Need to estimate & correct for the channel h



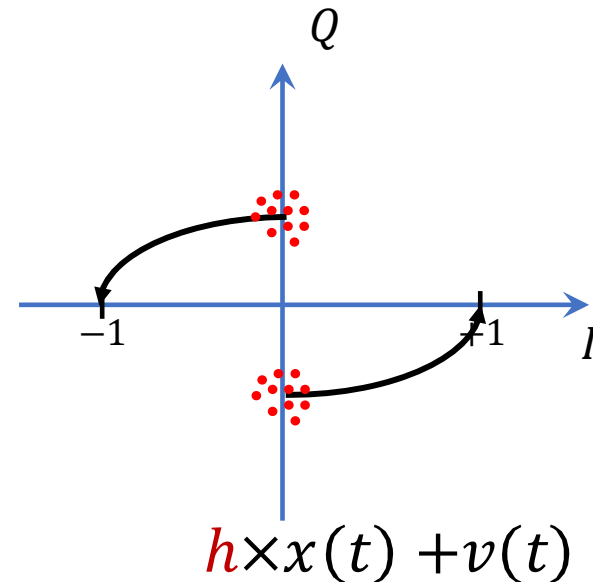
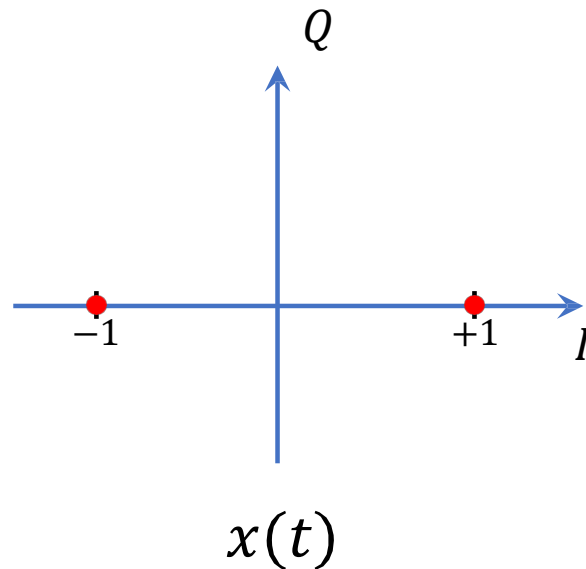
Channel Estimation & Correction

Channel Estimation & Correction

Consider BPSK Modulation.

$$0 \rightarrow -1$$

$$1 \rightarrow +1$$



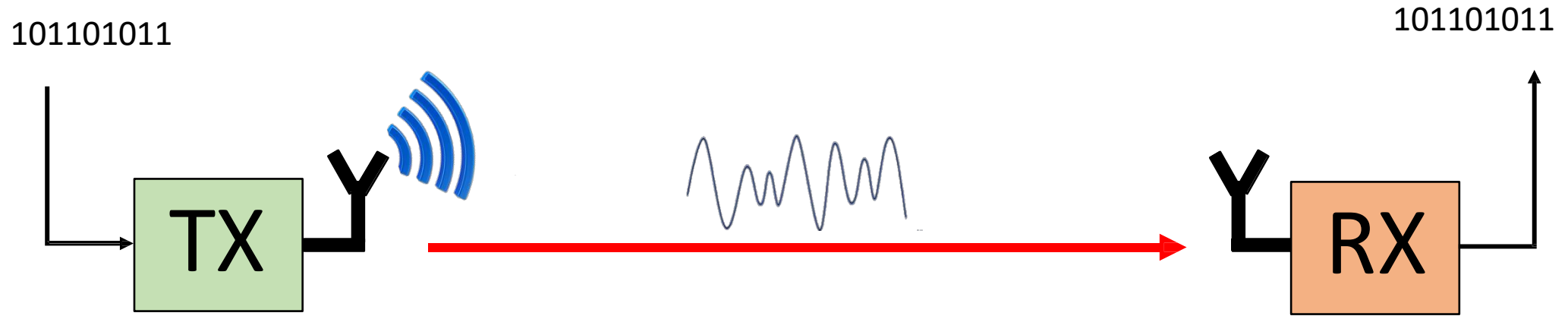
Send Training Sequence (Preamble Bits): Known Bits

$$\begin{aligned} x(0) = 1 &\longrightarrow y(0) = h + v(0) \\ x(1) = 1 &\longrightarrow \underline{y(1)} = h + v(1) \\ x(2) = -1 &\longrightarrow y(2) = -h + v(2) \end{aligned}$$

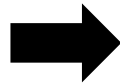
•
•
•

$$\begin{aligned} \text{Estimate channel: } \underline{\tilde{h}} &= \frac{1}{K} \sum_{k=1}^K \frac{y(k)}{x(k)} \\ \text{Correct channel: } \underline{\tilde{x}(t)} &= \frac{y(t)}{\tilde{h}(t)} \end{aligned} \quad \left. \begin{array}{l} \\ \text{MLE} \end{array} \right\}$$

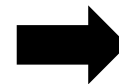
The Channel



$x(t)$



Channel

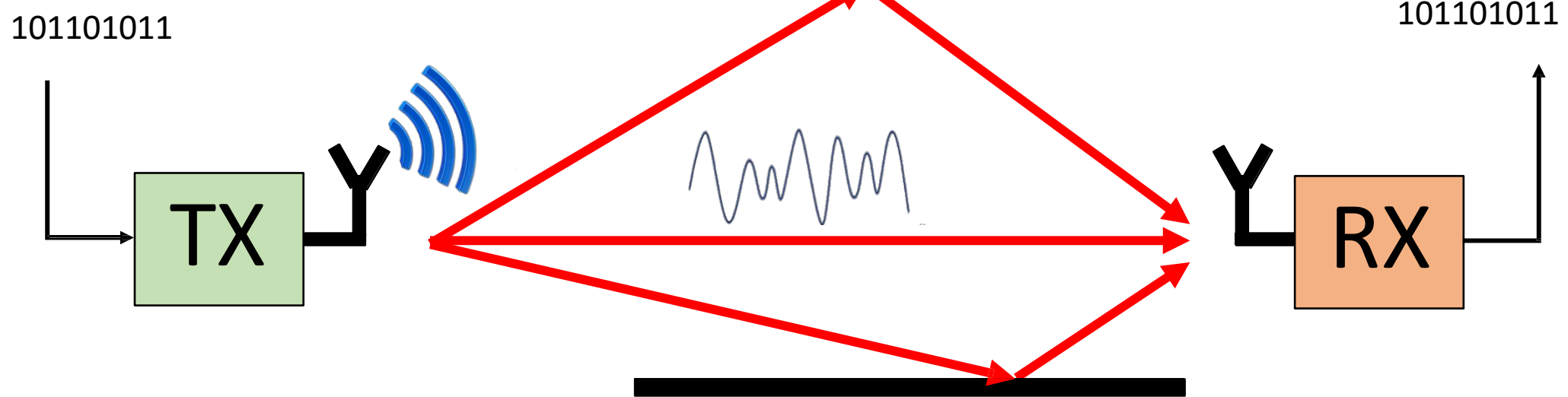


$y(t) = \mathbf{h} x(t - \tau) + v(t)$

$$h \propto \frac{\lambda}{d} e^{j2\pi d/\lambda}$$

Assumes single path!

Multipath Channel



Multipath Propagation: radio signal reflects off objects ground, arriving at destination at slightly different times

$$y(t) = \underbrace{\alpha_1 e^{j\phi_1}}_{\text{amplitude and phase}} \underbrace{x(t - \tau_1)}_{\text{delay}} + \underbrace{\alpha_2 e^{j\phi_2}}_{\text{amplitude and phase}} \underbrace{x(t - \tau_2)}_{\text{delay}} + \underbrace{\alpha_3 e^{j\phi_3}}_{\text{amplitude and phase}} \underbrace{x(t - \tau_3)}_{\text{delay}} \dots$$

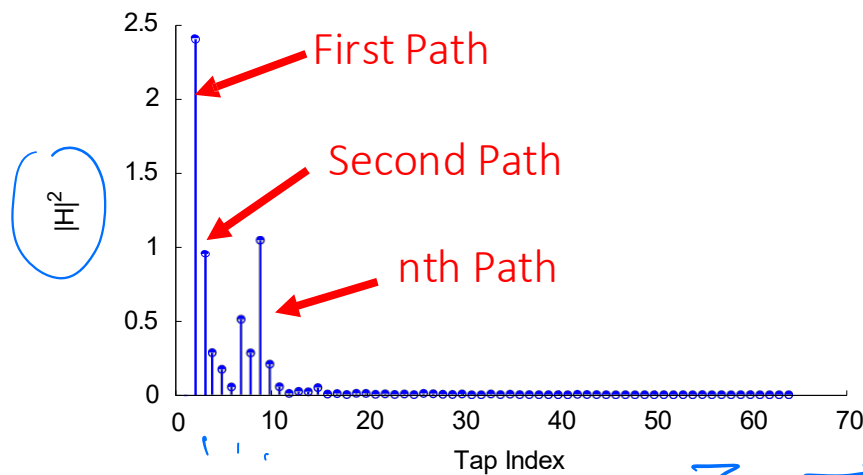
$$y(t) = \underbrace{\sum_k \alpha_k e^{j\phi_k}}_{\text{amplitude and phase}} \underbrace{x(t - \tau_k)}_{\text{delay}} = \underbrace{\sum_k h_k(\tau_k)}_{\text{channel impulse response}} \underbrace{x(t - \tau_k)}_{\text{delay}} = \underbrace{h(t) * x(t)}_{\text{convolution}}$$

$h(t)$ is channel impulse response.

Multipath Channel

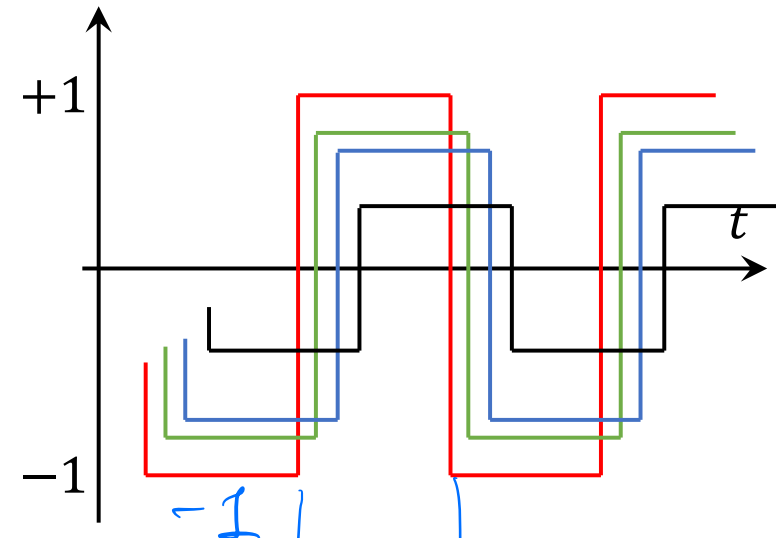
$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



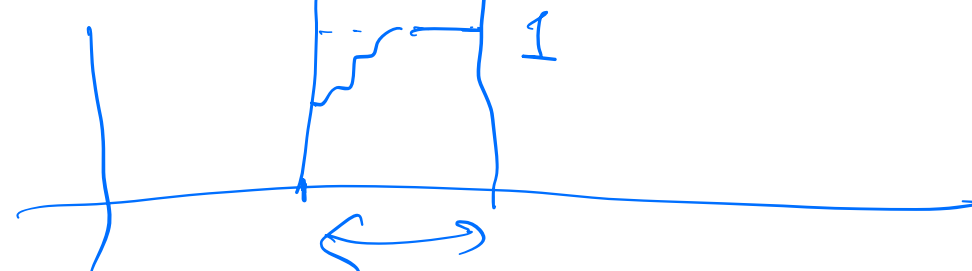
Multi-tap Channel

channel impulse response



ISI: Inter-Symbol-Interference

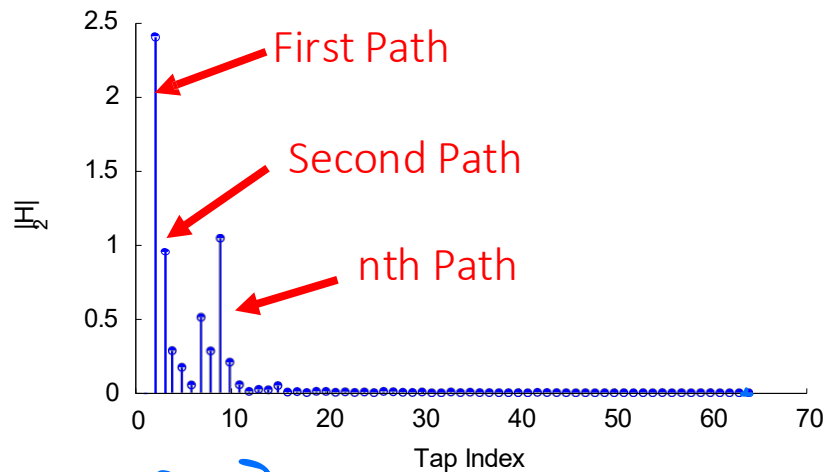
Symbols arriving along late paths interfere with following symbols.



Multipath Channel

$h(t)$ is channel impulse response.

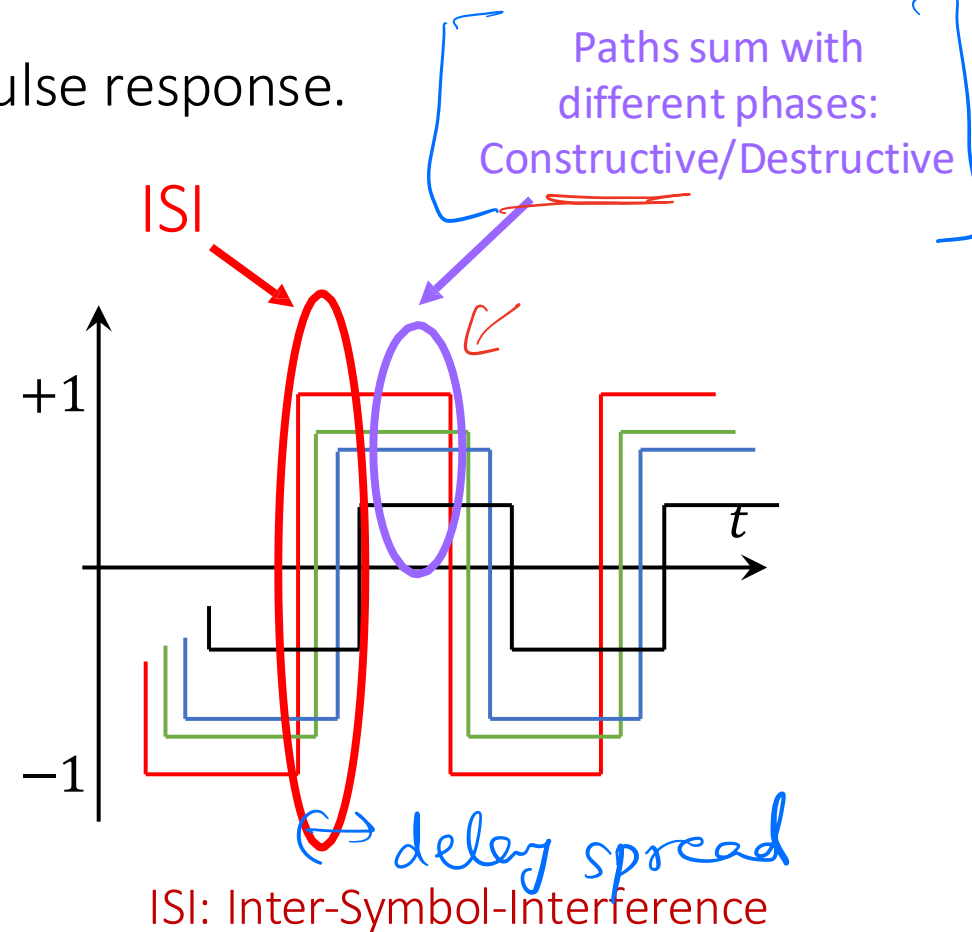
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$



Multi-tap Channel

delay spread

$\approx$ ms

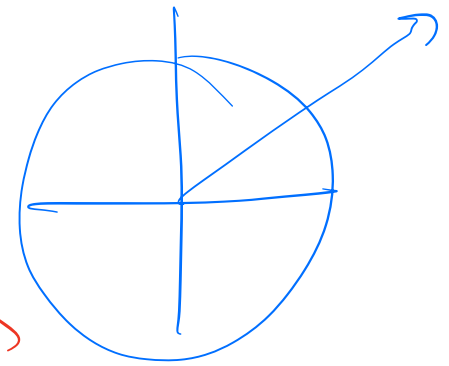
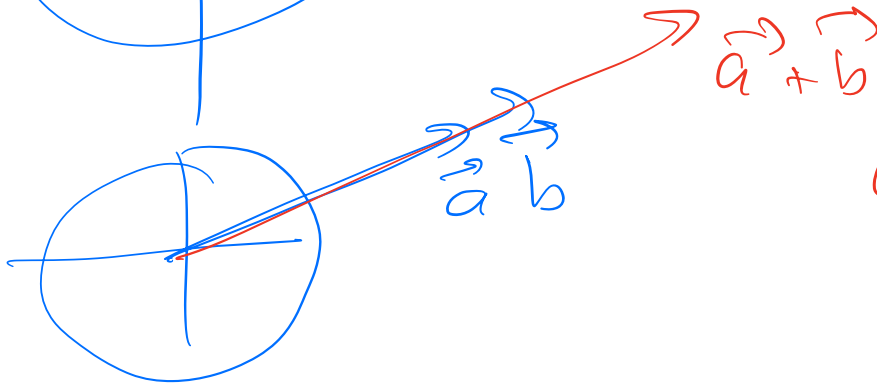
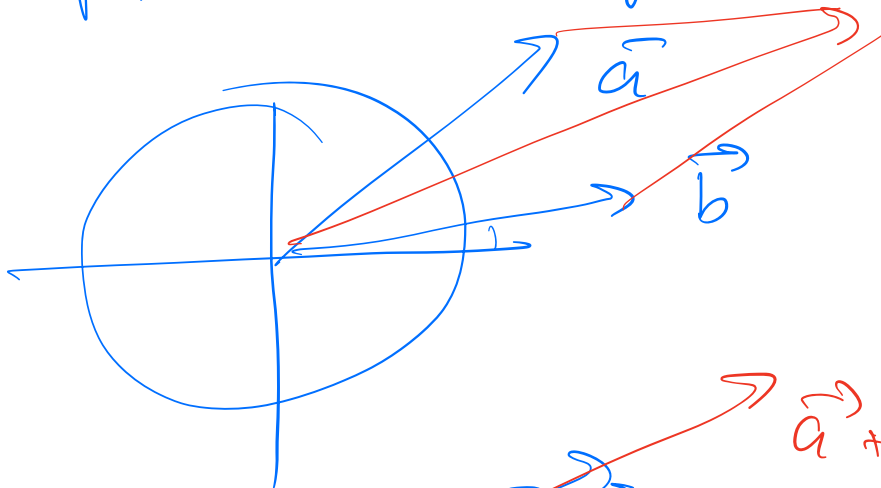


Symbols arriving along late paths interfere with following symbols.

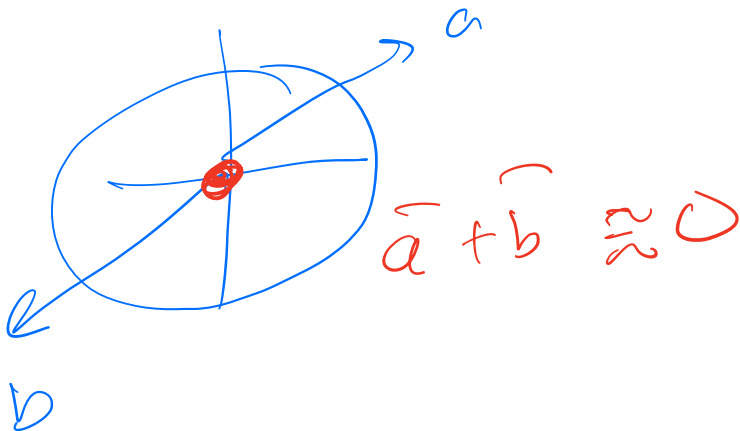
Channel Fading

Symbols arriving along different paths sum up destructively

Phaser diagram



constructive



$$\vec{a} + \vec{b} \approx 0$$

destructive
interference

x

Multipath Channel

$h(t)$ is channel impulse response.

$$\underline{y(t)} = \sum_k \underline{h(\tau_k)} \underline{x(t - \tau_k)} = \underline{h(t) * x(t)}$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m, d_2 = 1.06m$:

$$\begin{aligned} h &= \underline{h_1 + h_2} = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda} \\ &= \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} \left(1 + \frac{d_1}{d_2} e^{j2\pi (d_2 - d_1)/\lambda} \right) \quad \frac{d_1}{d_2} \approx 1 \end{aligned}$$

$$\text{if } \underline{\frac{d_2 - d_1}{\lambda} \approx \frac{1}{2}} \rightarrow h = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} (1 + \underbrace{e^{j\pi}}_{-1}) = 0 \quad \text{Destructive Interference!}$$

Multipath Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

Channel Fading: Symbols arriving along different paths sum up destructively

Example 2 paths with distance $d_1 = 1m, d_2 = 1.06m$:

$$h = h_1 + h_2 = \frac{\lambda}{d_1} e^{j2\pi d_1/\lambda} + \frac{\lambda}{d_2} e^{j2\pi d_2/\lambda}$$

$$@f_1 = 2.5GHz (\lambda = 12 cm): \quad h = 0.12 e^{j\frac{2\pi}{3}} + 0.113 e^{j\frac{5\pi}{3}} \approx 0.006$$

$$@f_2 = 5GHz (\lambda = 6 cm): \quad h = 0.06 e^{j\frac{5\pi}{3}} + 0.05 e^{j\frac{5\pi}{3}} \approx 0.116$$

17×

(24dB)

Frequency Selective Fading

Multipath Channel

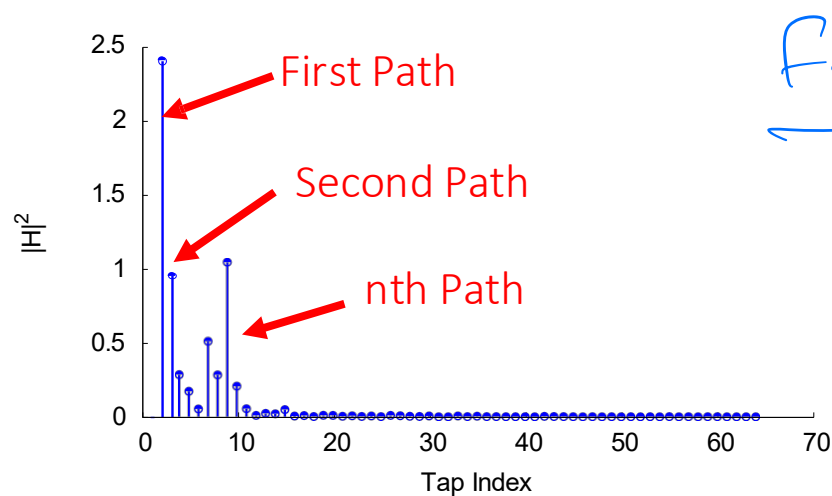
$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

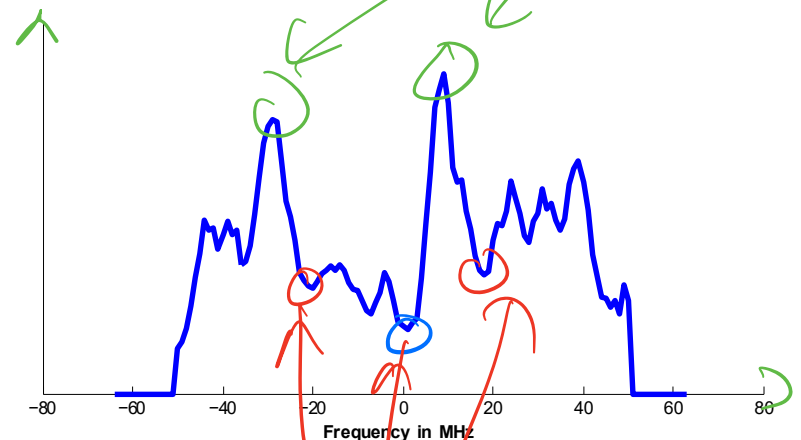
$\Leftrightarrow$

$$H(f)X(f)$$

constructive addition.



Multi-tap Channel



f

$$h \propto \delta(t - \tau) \xrightarrow{\text{FFT}} 1 e^{j2\pi f \tau}$$

$$h_1 \delta(t - \tau_1) + h_2 \delta(t - \tau_2) \dots$$

$$\rightarrow h_1 e^{j2\pi f \tau_1} + h_2 e^{j2\pi f \tau_2} \dots$$

Multipath Channel

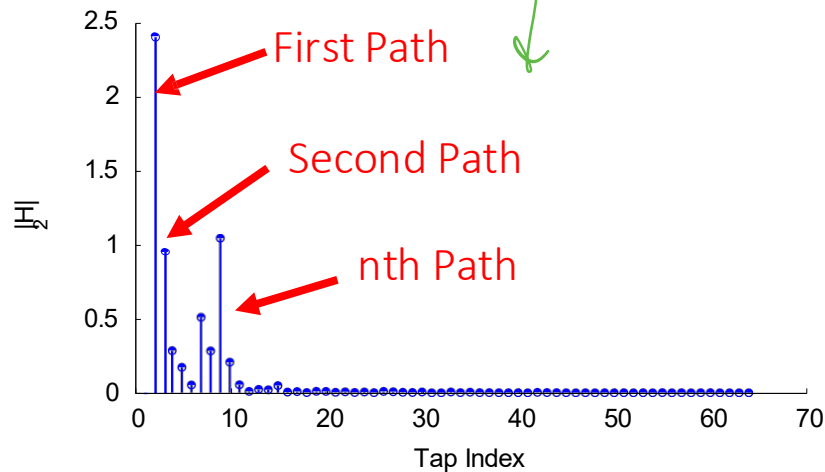
delay

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$\Leftrightarrow$

$$H(f)X(f)$$

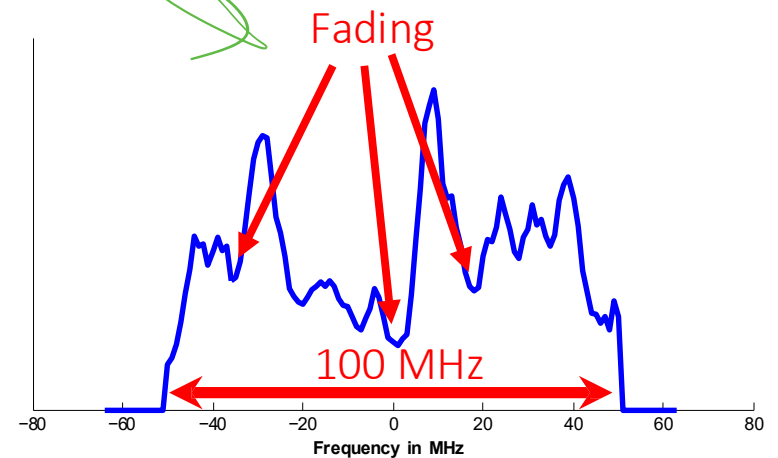


Multi-tap Channel



ISI: Inter-Symbol-Interference

Symbols arriving along late paths interfere with following symbols.



Frequency Selective Fading

Symbols arriving along different paths sum up destructively

Problematic in Wideband Channel!

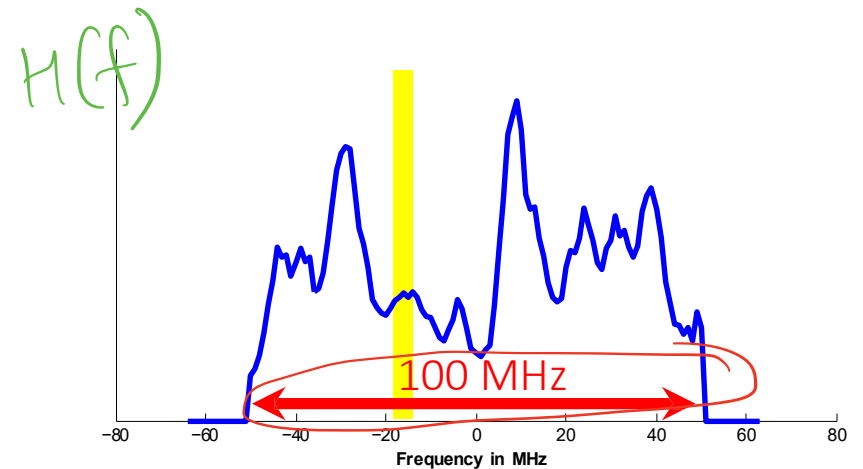
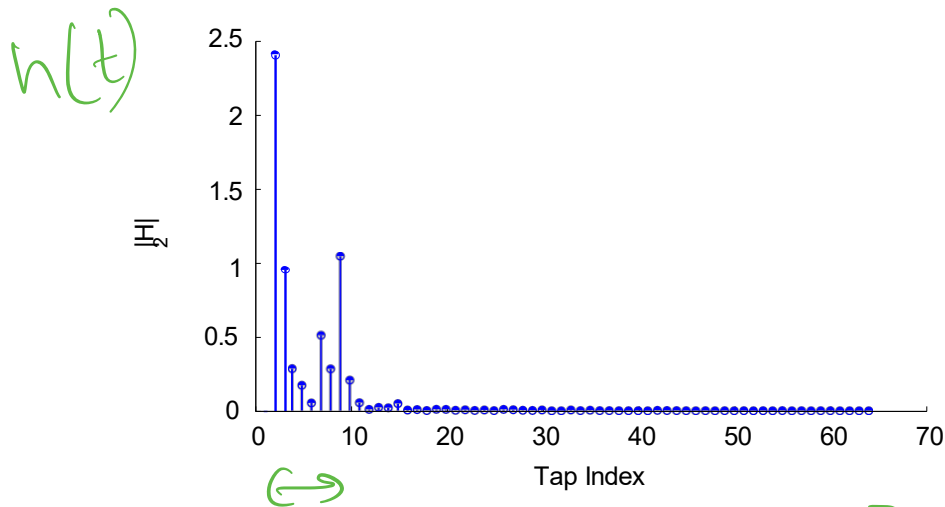
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = \underline{h(t) * x(t)}$$

$\Leftrightarrow$

$$H(f)X(f)$$



ISI

$\rightarrow z$

FSF

$\rightarrow f$

$x(f)$

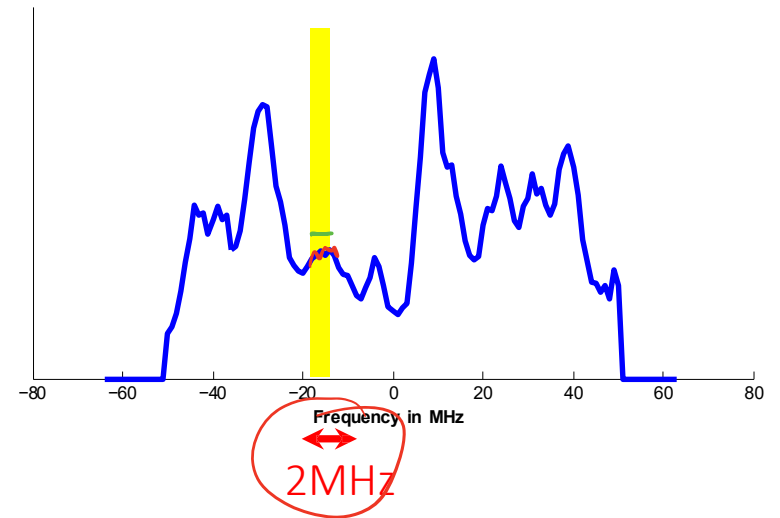
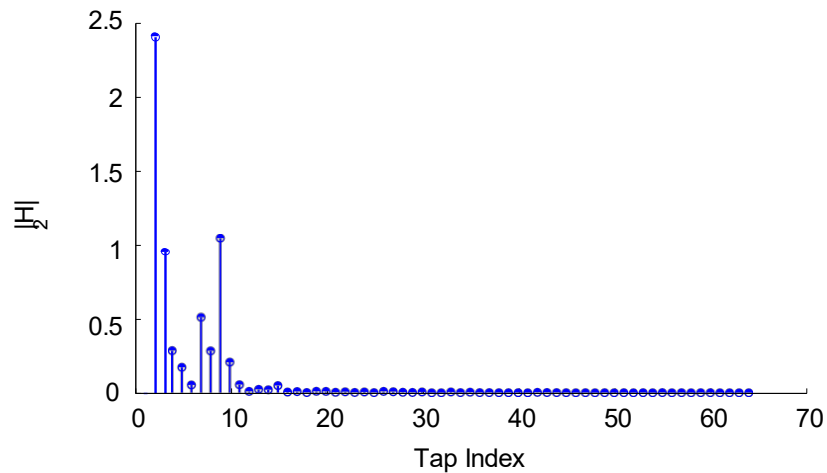
$x(f)H(f)$



Narrowband Channel

$h(t)$ is channel impulse response.

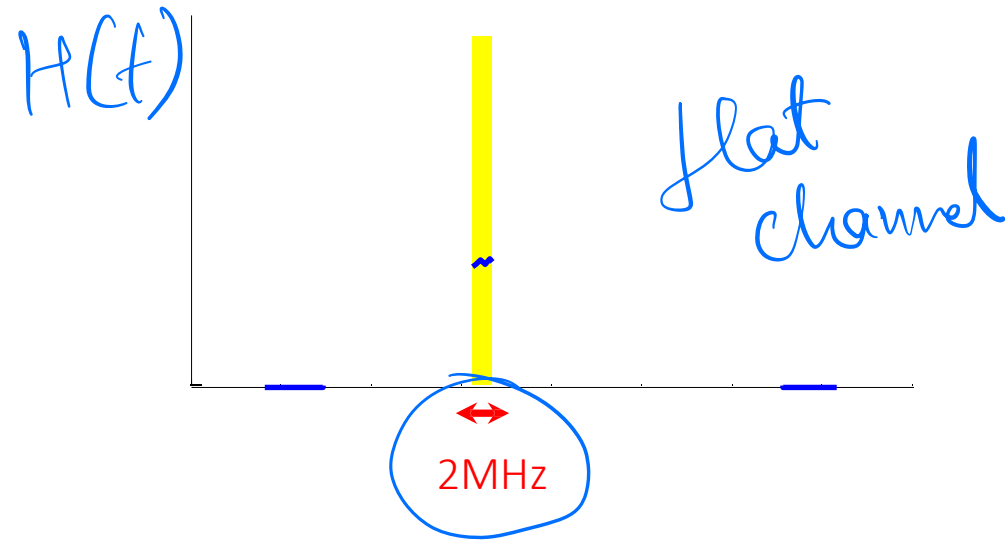
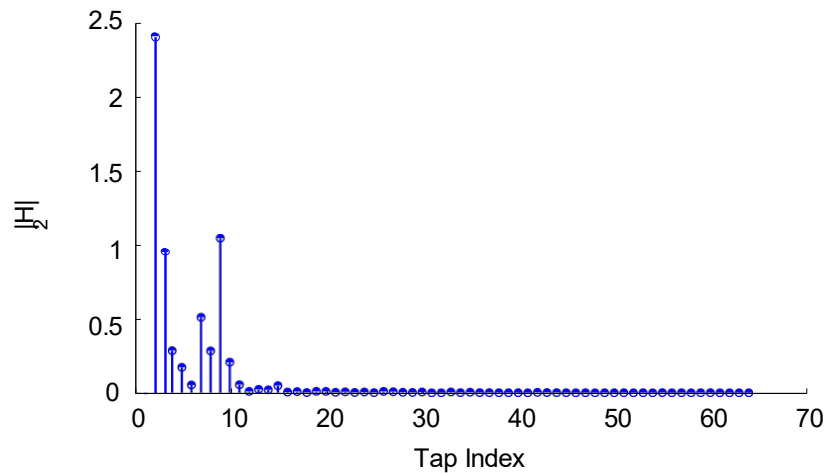
$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



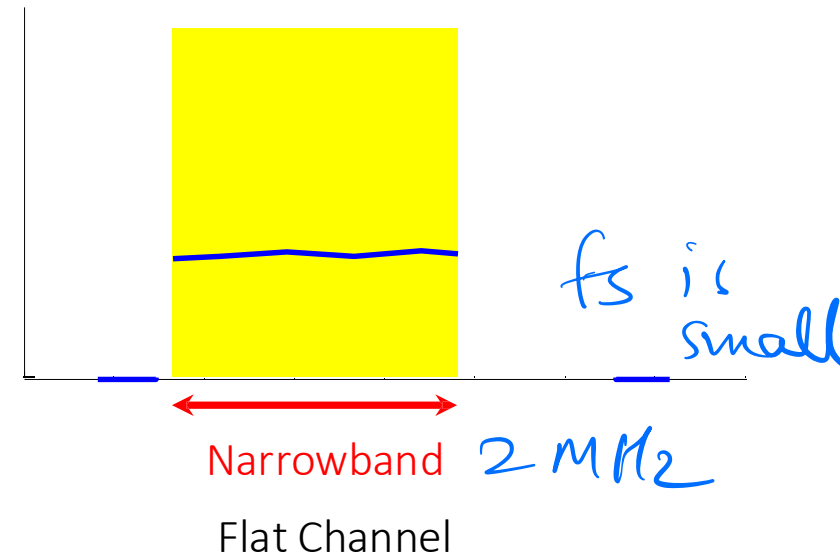
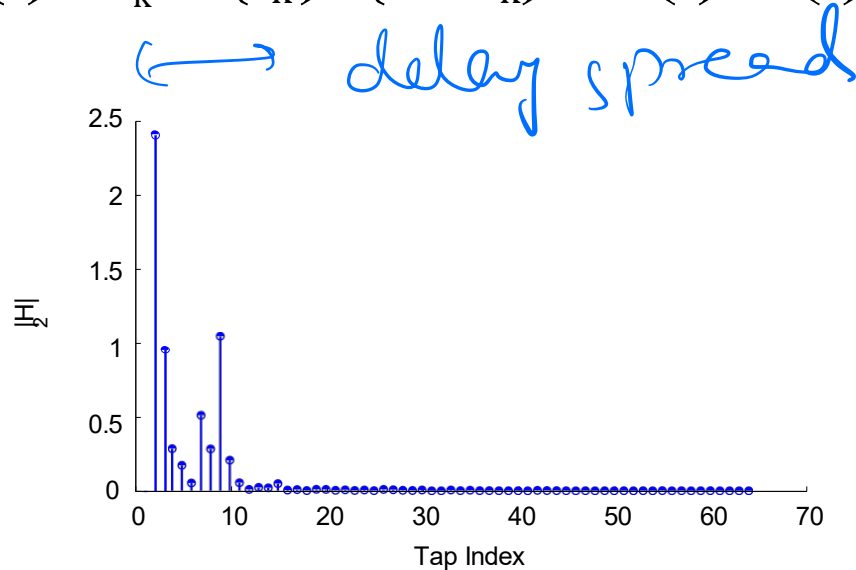
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$\Leftrightarrow$

$$H(f)X(f)$$

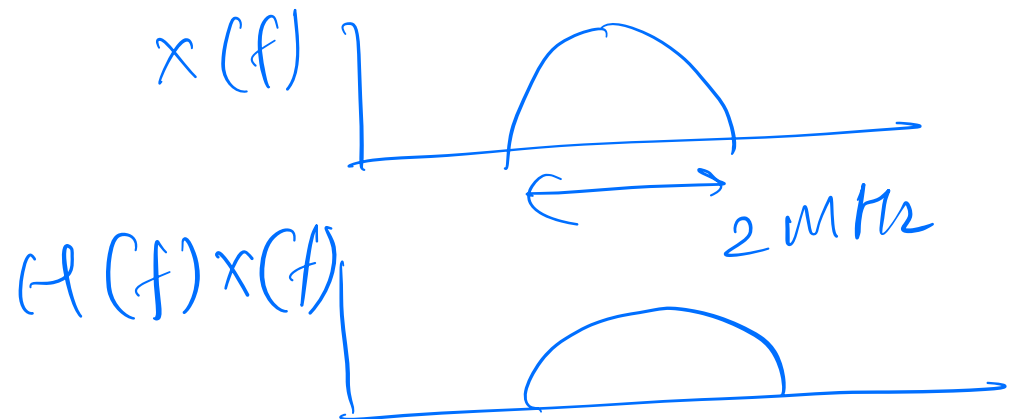


Symbol time: $T \propto \frac{1}{\text{bandwidth}}$

$\gg \tau_k$

$$\tau_s \approx \frac{1}{f_s} = \frac{1}{2M}$$

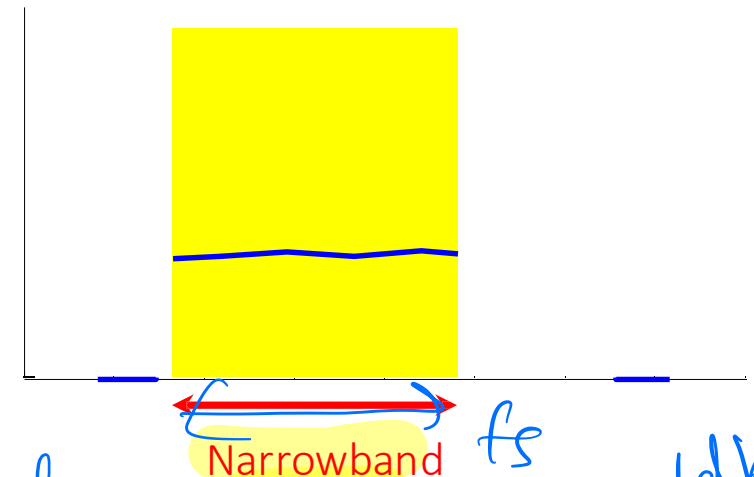
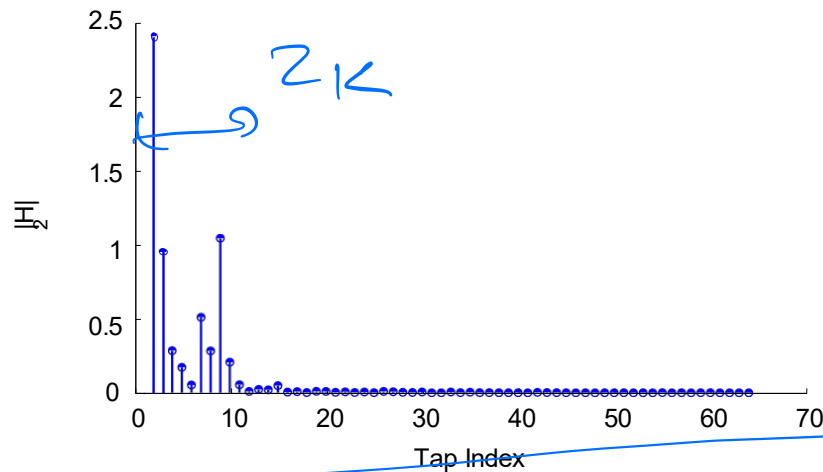
$$\approx 5 \mu s$$



Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t) \quad \Leftrightarrow \quad H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$

$\gg \tau_k$

delay spread

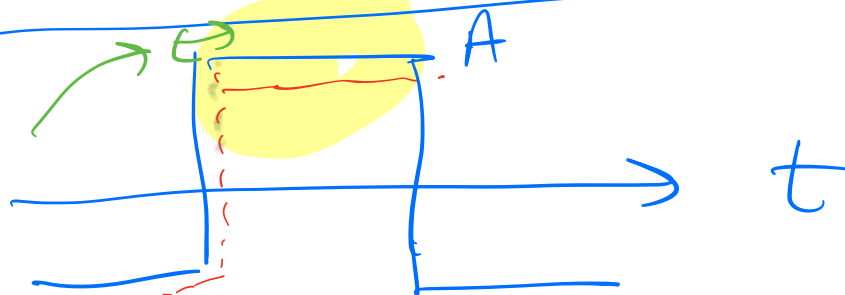
Narrowband
Flat Channel

should be large

$T_s \gg Z_k$
 $\Rightarrow$ No ISI

$$f_s = \frac{1}{T_s}$$

delay τ_k



$-A \leftarrow T_s \leftarrow$ symbol period

Narrowband Channel

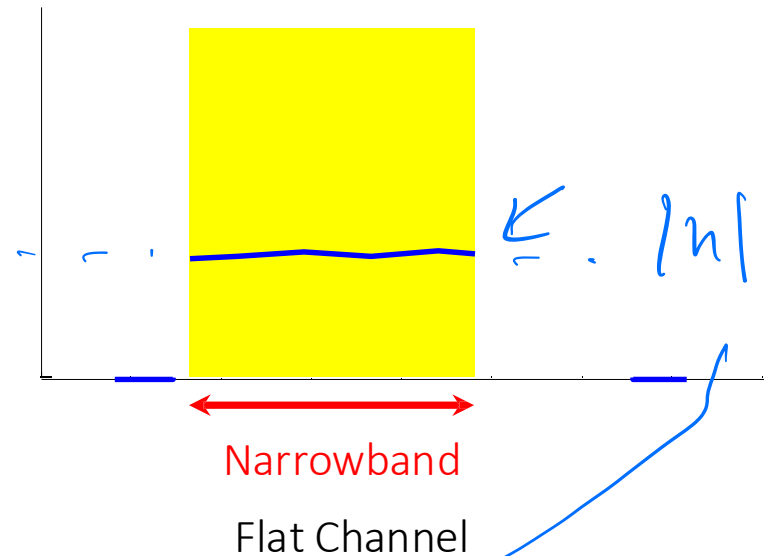
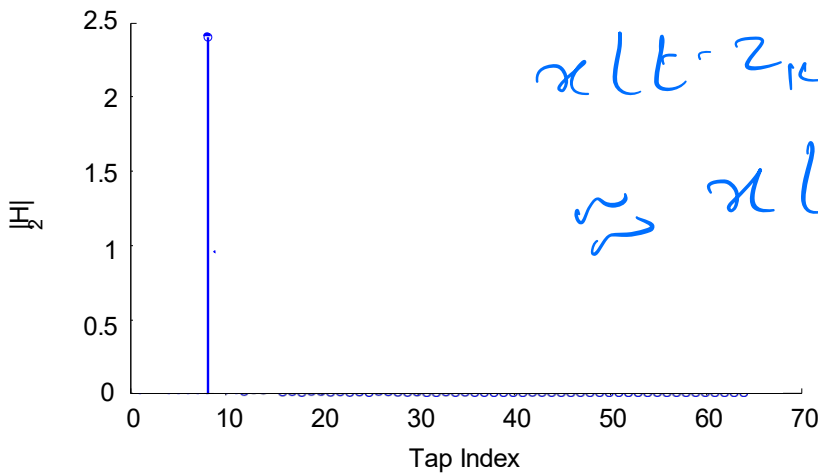
$h(t)$ is channel impulse response.

← freq response of
channel is flat
over narrowband
bandwidth

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) = h(t) * x(t)$$

$\Leftrightarrow$

$$H(f)X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$

$\gg \tau_k$

$h(t) * x(t)$

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = hx(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

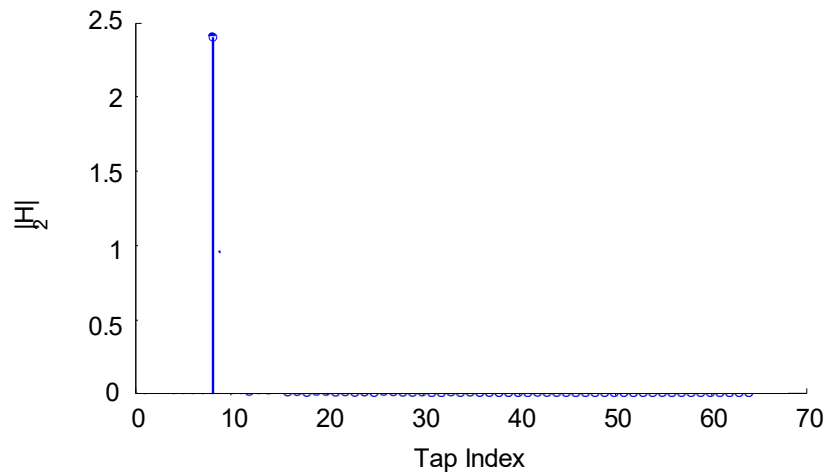
$y(t) = hx(t)$

Matlab

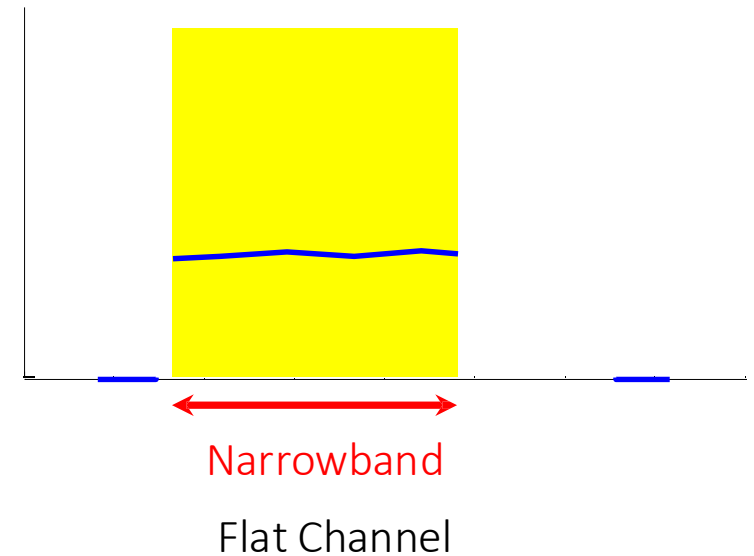
Narrowband Channel

$h(t)$ is channel impulse response.

$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx h x(t) \quad \Leftrightarrow \quad h X(f)$$



Symbol time: $T \propto \frac{1}{\text{bandwidth}}$ $\gg \tau_k$



$$y(t) = \sum_k h(\tau_k) x(t - \tau_k) \approx \sum_k h(\tau_k) x(t) = \left(\sum_k h(\tau_k) \right) x(t) = h x(t)$$

Narrowband Channel is Approximated by a Single Tap Channel

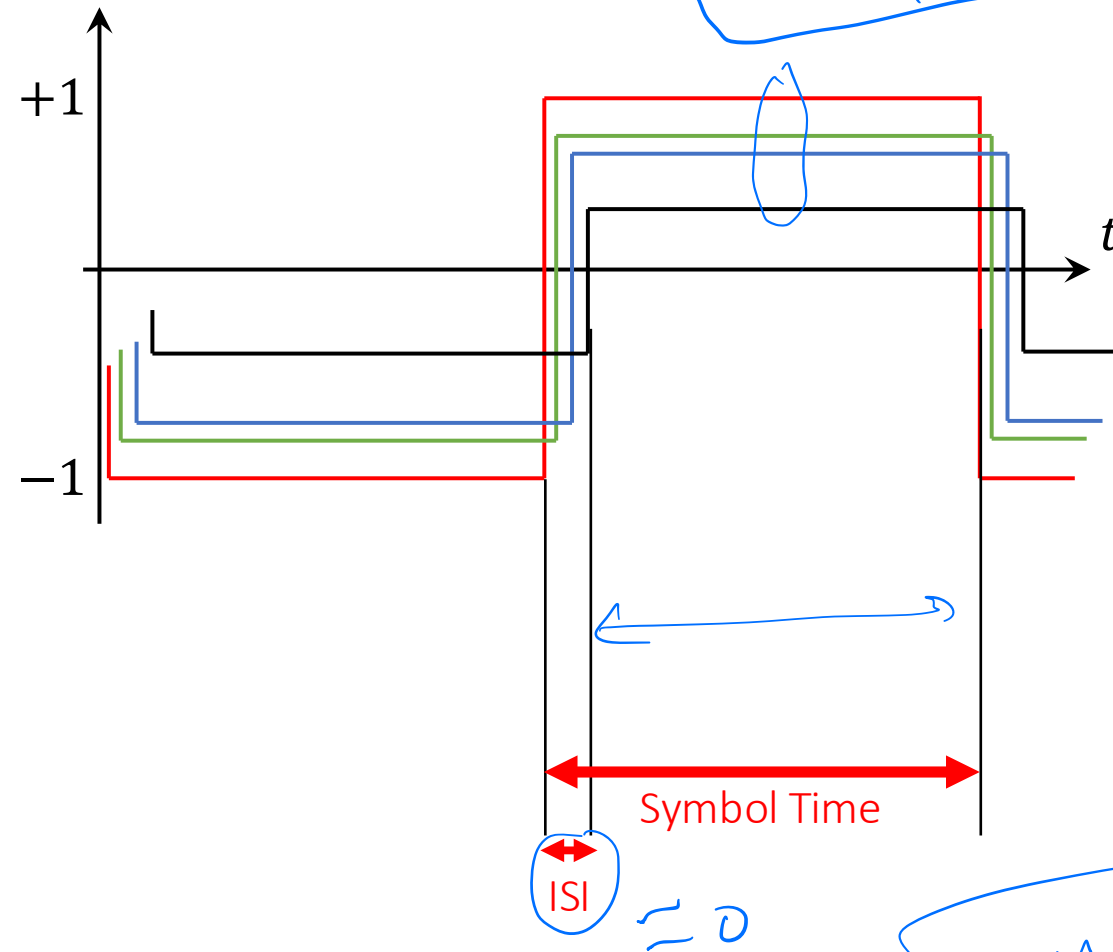
Narrowband vs. Wideband Channel

time
domain

Narrowband Channel

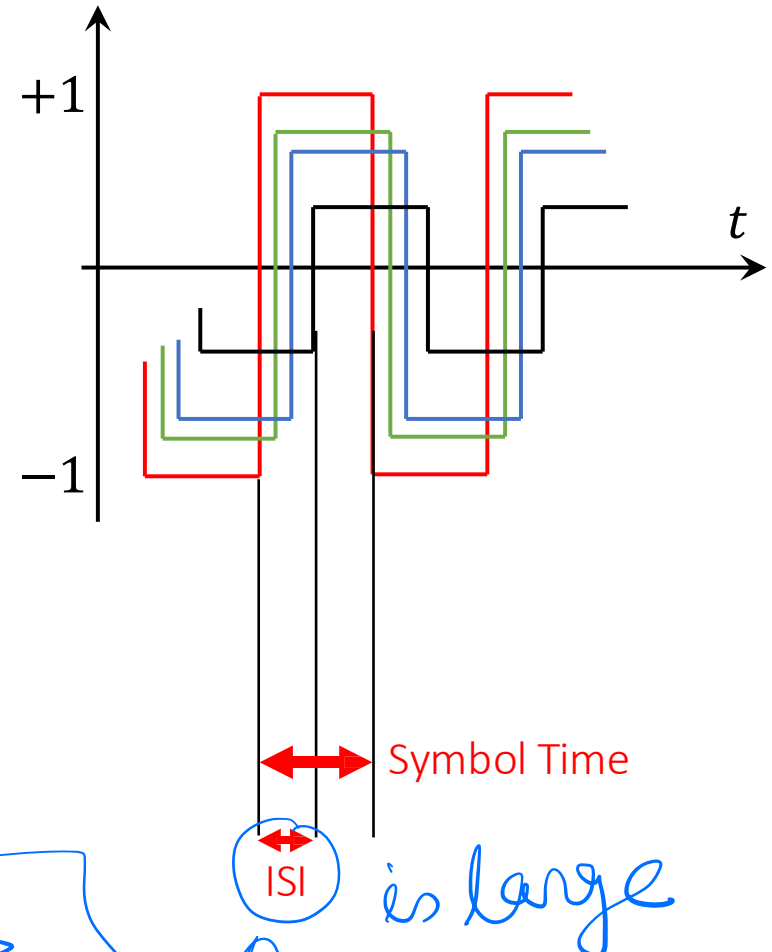
$h x(t)$

$h = \sum_k h_k$



Wideband Channel

$h(t) * x(t)$



TAKES
model

is large

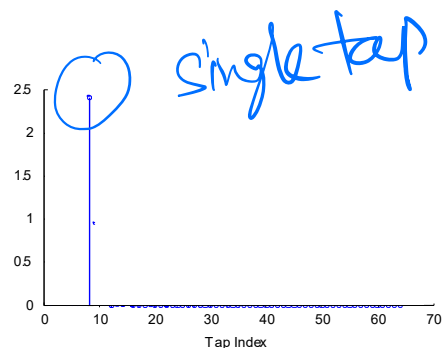
Narrowband vs. Wideband Channel

Narrowband Channel

$$h x(t)$$

$$\Leftrightarrow$$

$$h X(f)$$



$\approx$ Single Tap



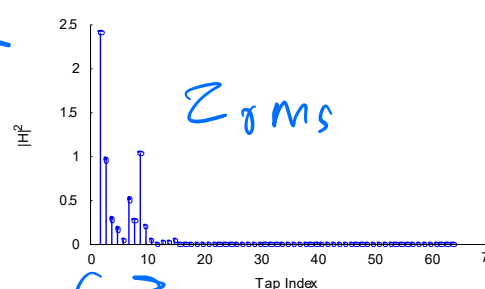
$\approx$ Flat Channel

Wideband Channel

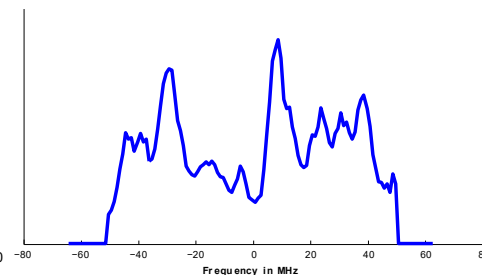
$$h(t) * x(t)$$

$$\Leftrightarrow$$

$$H(f) X(f)$$



Multi-tap
Channel
(ISI)

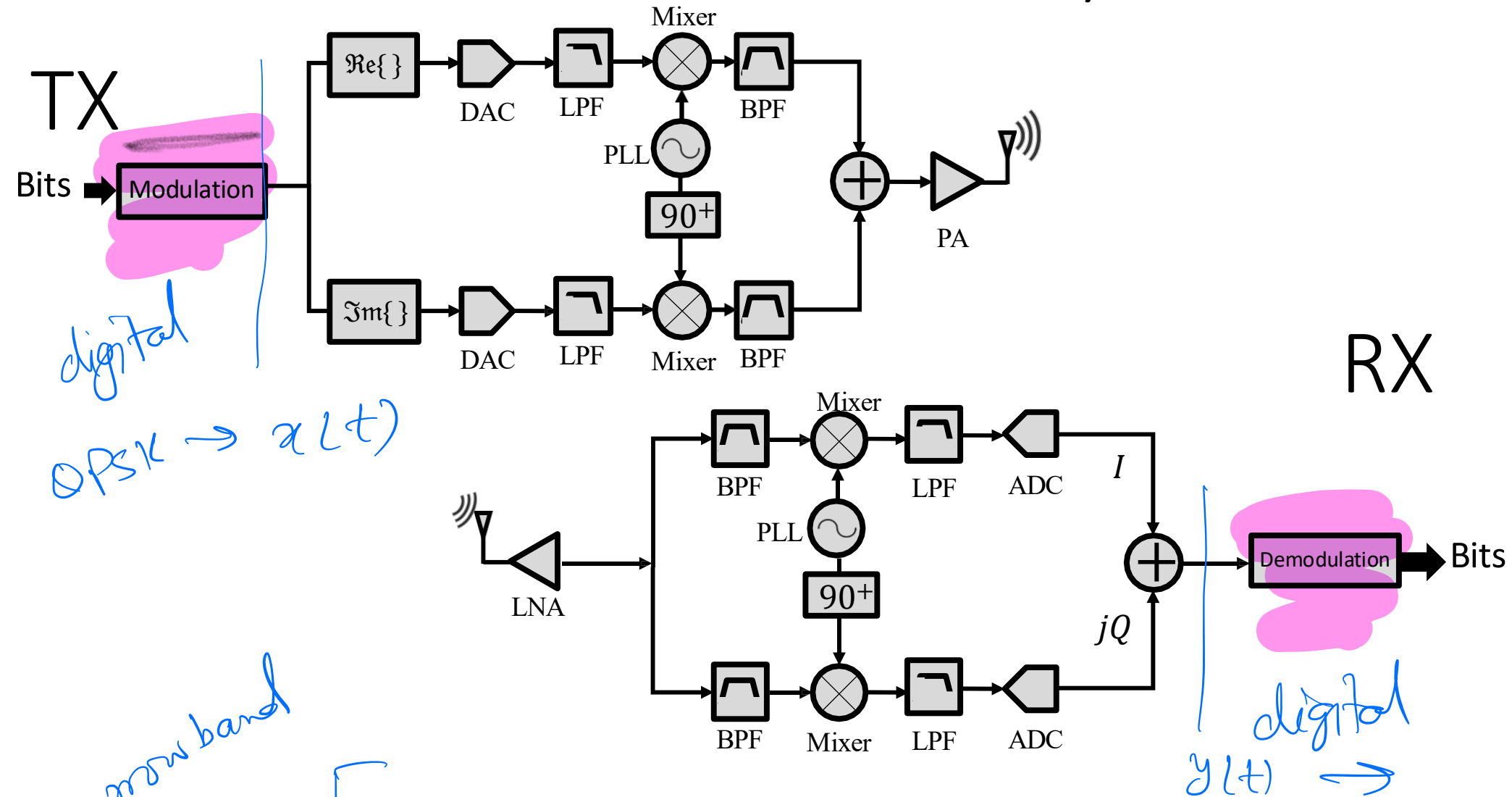


Frequency
Selective Channel

Need to correct for ISI to
be able to decode
correctly!

Estimating and Correcting for multi-tap channel is hard
Simplify processing using: **OFDM**

Wireless Communication System



Single Carrier Modulation

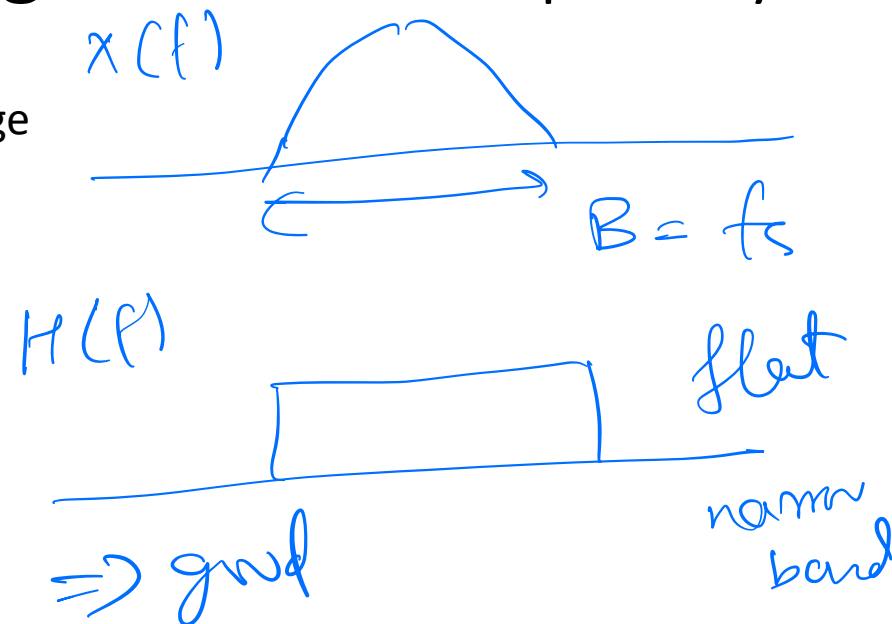
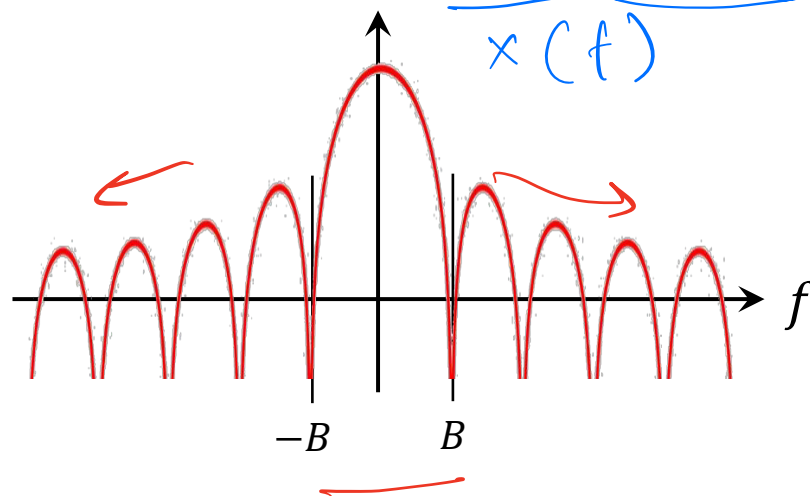
Symbols modulated on a single carrier frequency: $s[n]e^{-j2\pi f_c t}$

Re

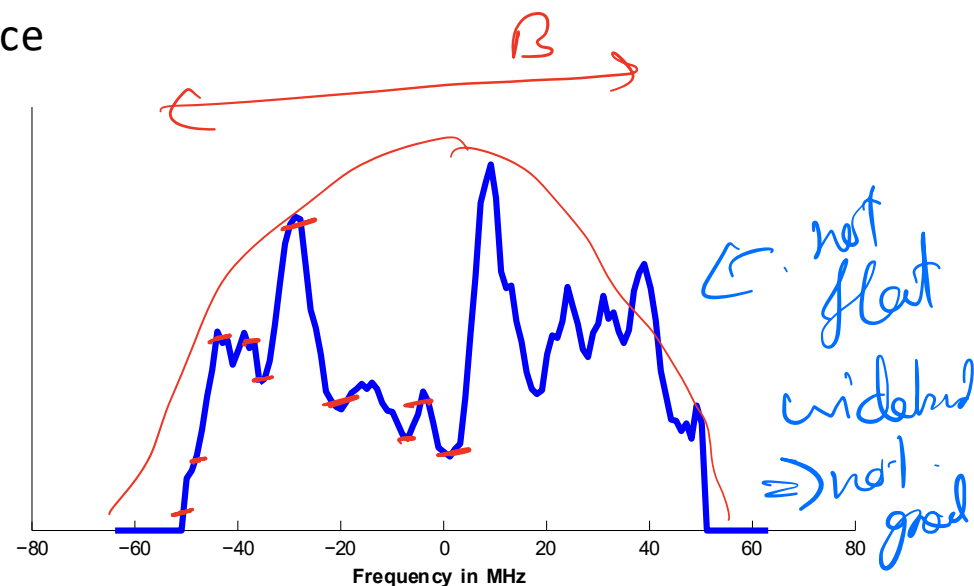
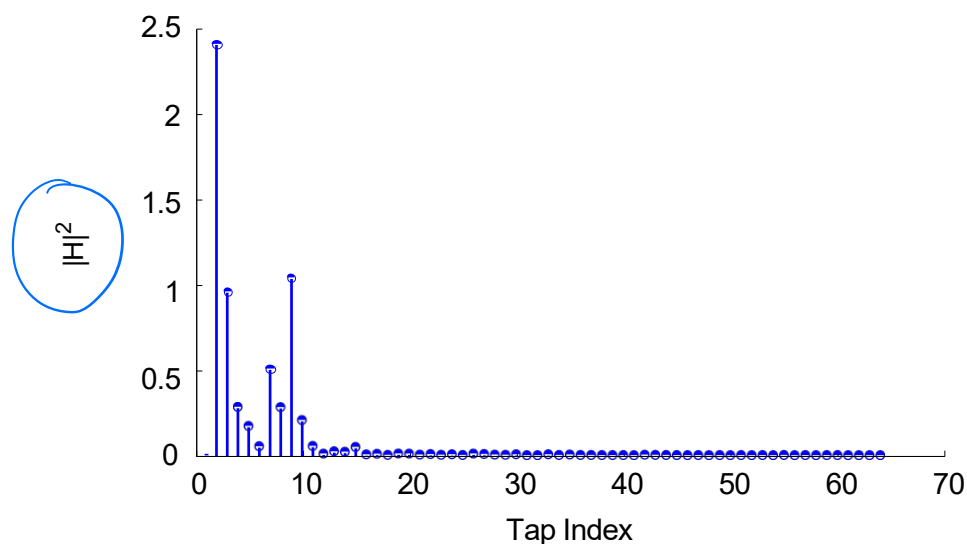
Single Carrier Modulation

Symbols modulated on a single carrier frequency

- Low Spectral Efficiency: sinc & raised cosine leakage



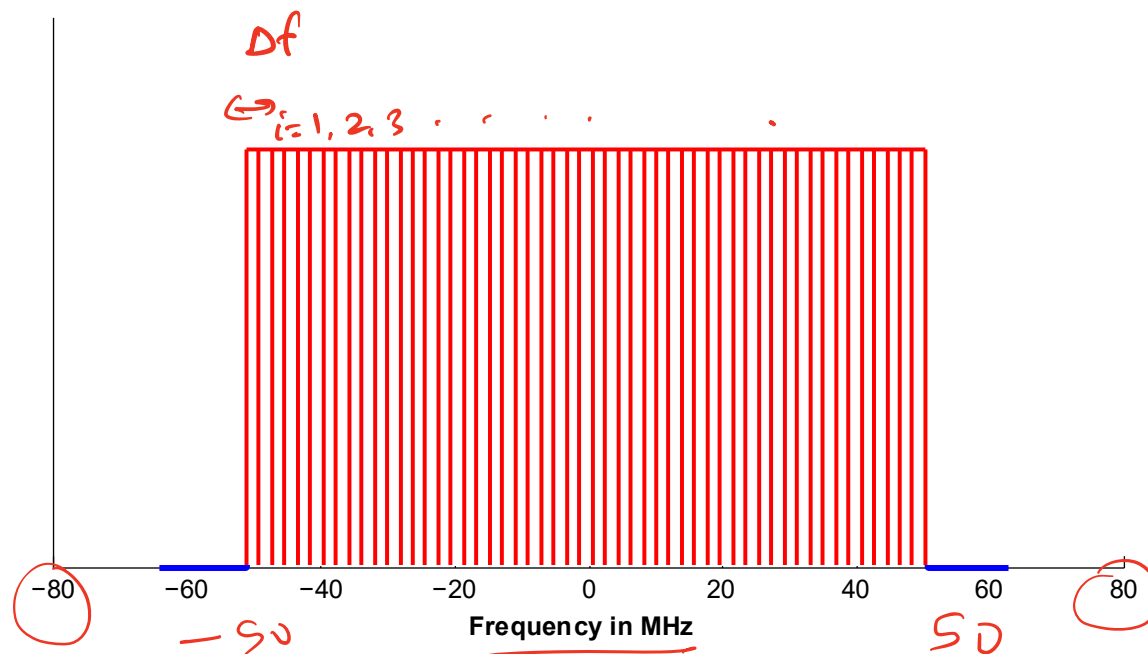
- ISI: Inter-Symbol-Interference limits performance



Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

$$x(t) = \sum_i s_i[n] e^{-j2\pi f_i t}$$

$s_i[n] \sum_i e^{-j2\pi f_i t}$

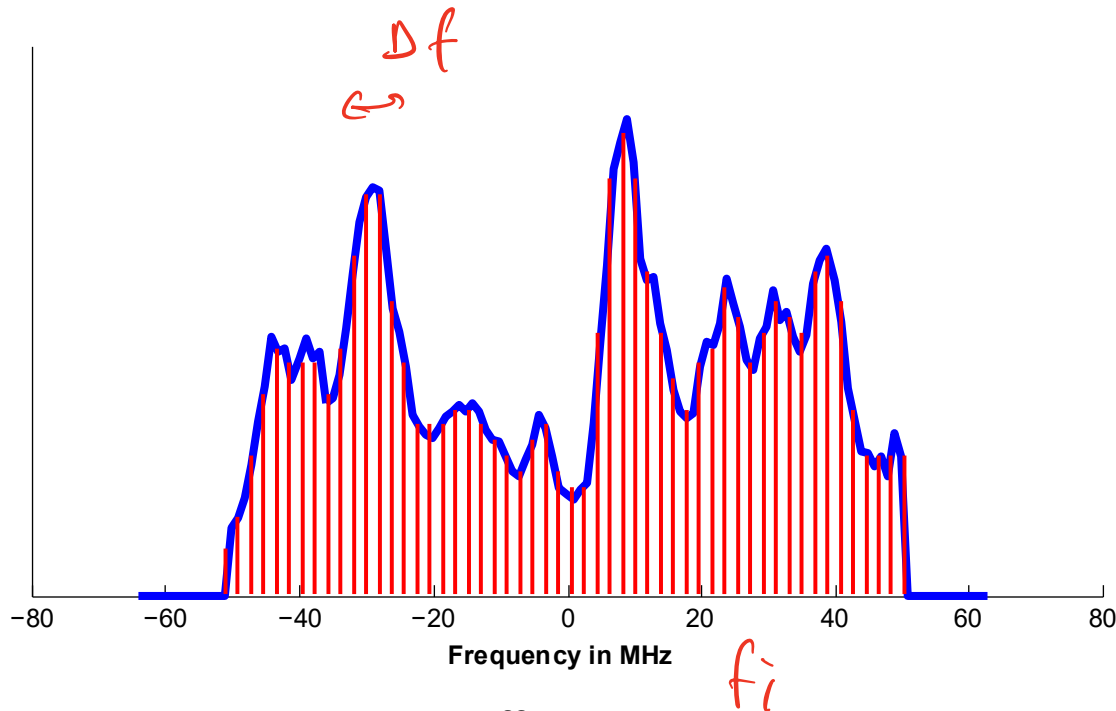
$S_i(f)$

A hand-drawn diagram showing three bell-shaped curves representing the frequency response $S_i(f)$ of individual sub-carriers. Each curve is centered at a specific frequency, with arrows pointing to these centers labeled f_1 , f_2 , and f_3 . Below the diagram, there is a large red double-headed arrow.

Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



$$x(t) = \sum_i s_i[n] e^{-j2\pi f_i t}$$

$$y(t) = \sum_i h_i s_i[n] e^{-j2\pi f_i t}$$

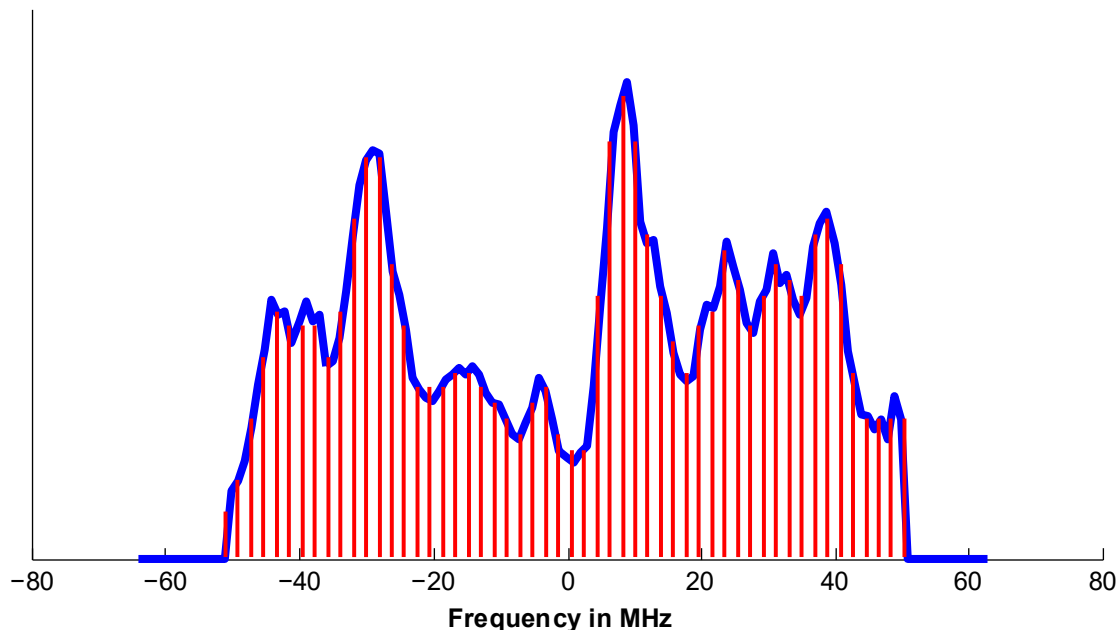
h_i = channel at f_i
 $\Rightarrow$ narrowband within Δf

- Transmit symbols on different carriers in narrow bands
- Channel is Flat $\rightarrow$ No need to worry about ISI

Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



$$x(t) = \sum_i s_i[n] e^{-j 2 \pi f_i t}$$

$$y(t) = \sum_i h_i s_i[n] e^{-j 2 \pi f_i t}$$

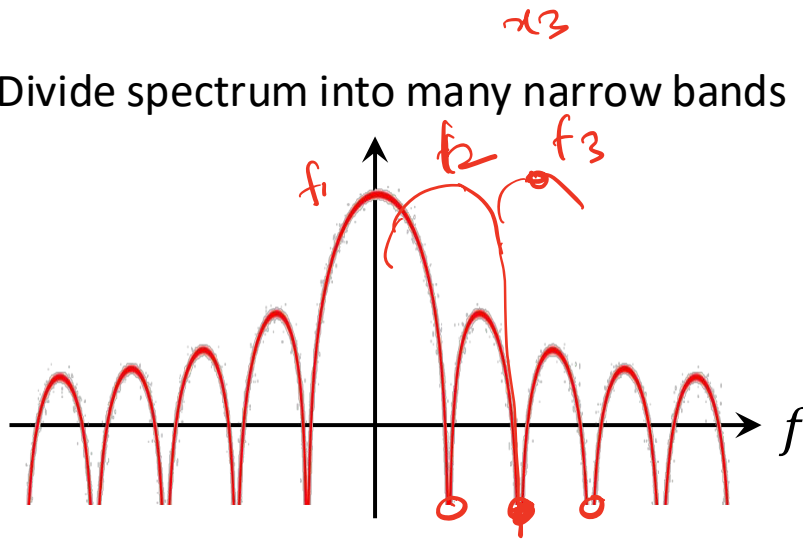
- Transmit symbols on different carriers in narrow bands
- Channel is Flat → No need to worry about ISI

Not That Simple!

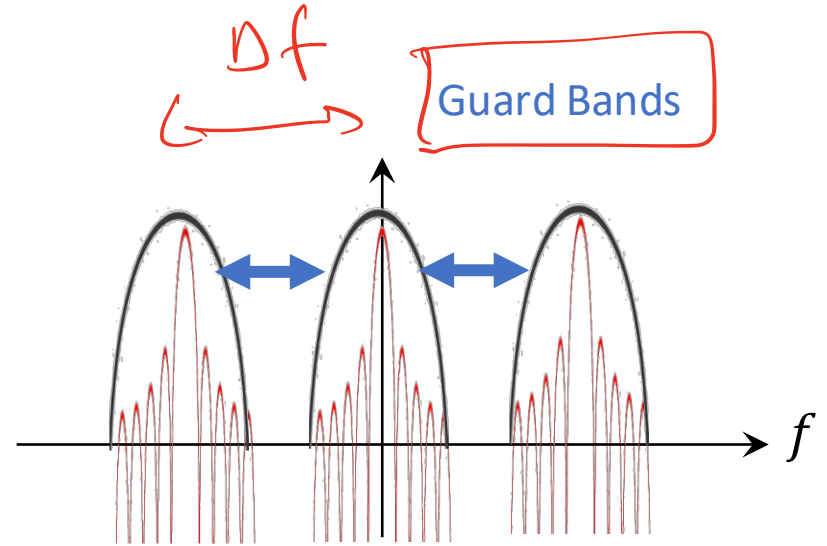
Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

- Divide spectrum into many narrow bands



- Significant Leakage between adjacent subcarriers
- Need Guard Bands → Very inefficient!

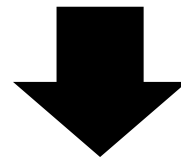
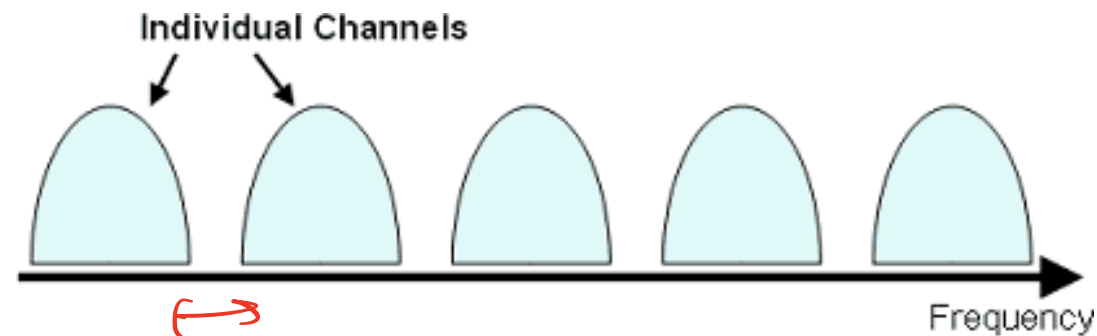


Solution: Make the Sub-Carriers Orthogonal

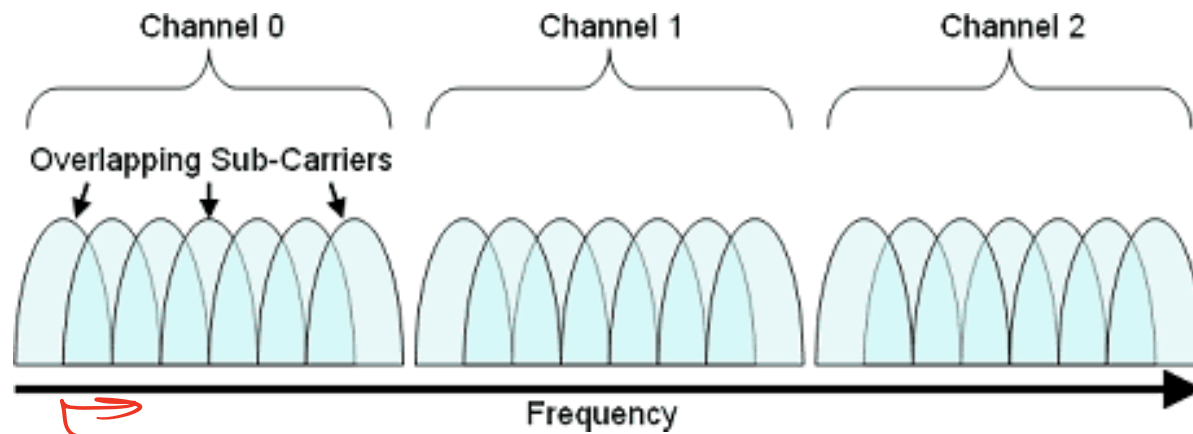
Multi-Carrier Modulation

Symbols modulated on multiple Sub-carrier frequencies

Make the Sub-Carriers Orthogonal



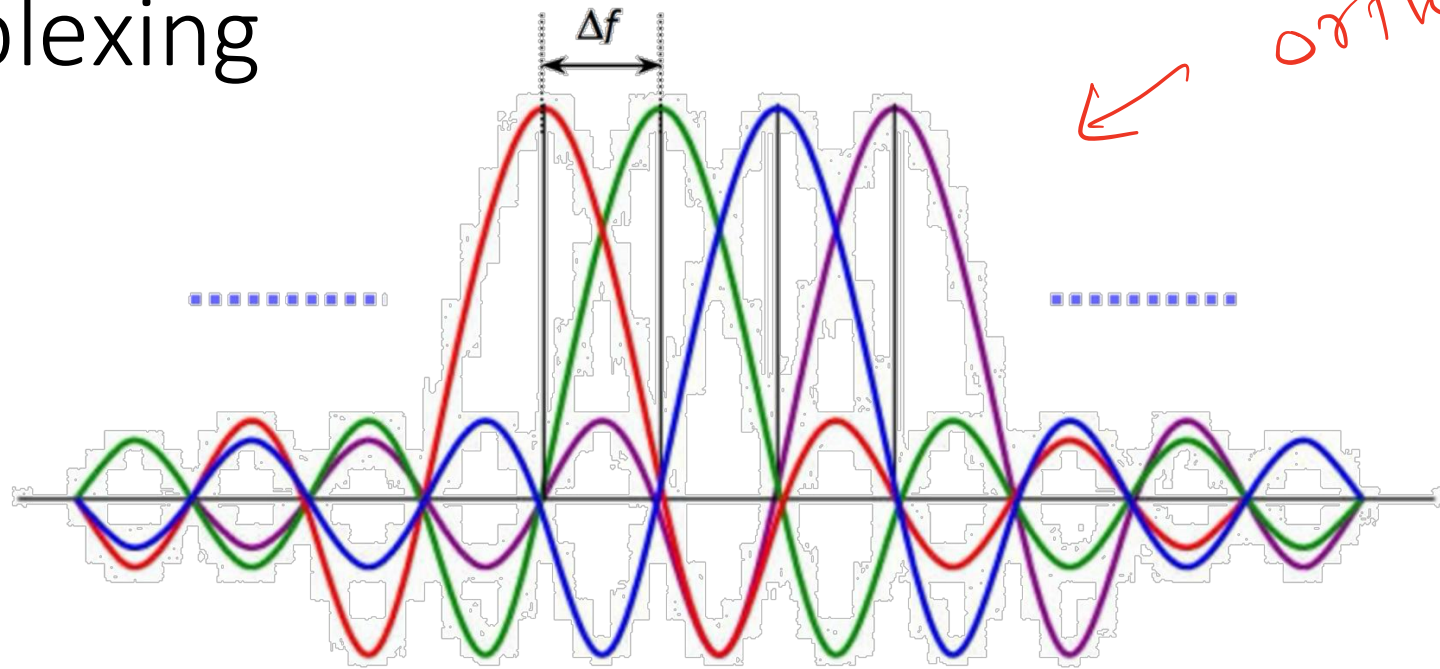
OFDM



OFDM
no guard bands

OFDM

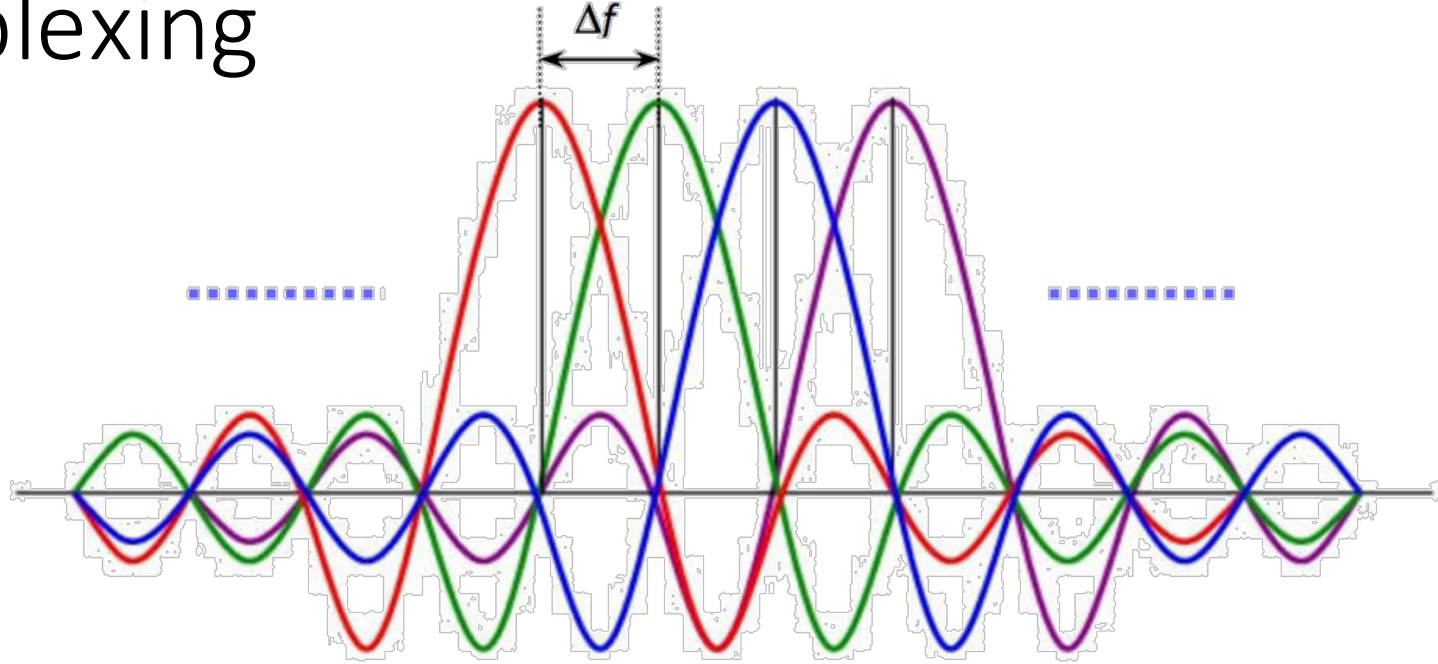
OFDM: Orthogonal Frequency Division Multiplexing



- Subcarriers are orthogonal: At the sub-carrier frequency, the sampled value has zero leakage from other subcarriers.
- Subcarrier separation can be very small, for N subcarriers and bandwidth B :

$$\Delta f = \frac{B}{N}$$

OFDM: Orthogonal Frequency Division Multiplexing



- Subcarriers are orthogonal: At the sub-carrier frequency, the sampled value has zero leakage from other subcarriers.
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$$\Delta f = \frac{B}{N}$$

How to Achieve This?

OFDM: Orthogonal Frequency Division Multiplexing

Use DFT: Discrete Fourier Transform

$$\text{N-Point DFT: } X(f_i) = \frac{1}{N} \sum_{t=0}^{N-1} x(t) e^{-j \frac{2\pi f_i t}{N}}$$

$$\text{N-Point IDFT: } x(t) = \sum_{f_i=0}^{N-1} X(f_i) e^{j \frac{2\pi f_i t}{N}}$$

Send symbols in Frequency Domain

$X(f_i) = s[n] \rightarrow$ Compute and transmit $x(t)$ using IDFT

OFDM: Orthogonal Frequency Division Multiplexing

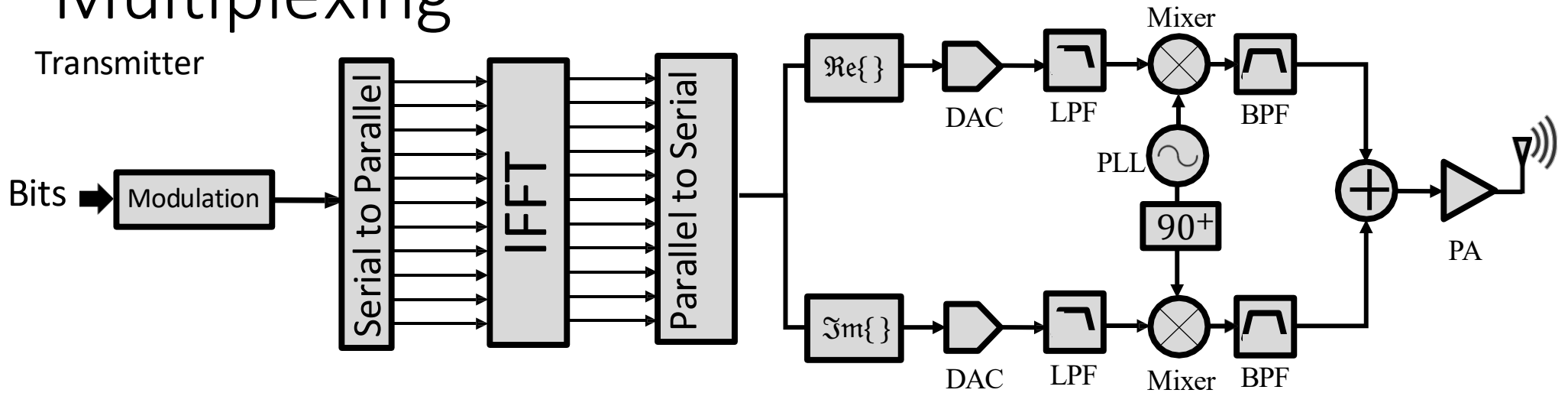
Send symbols in Frequency Domain

$X(f_Q) = s[n] \rightarrow$ Compute and transmit $x(t)$ using IDFT

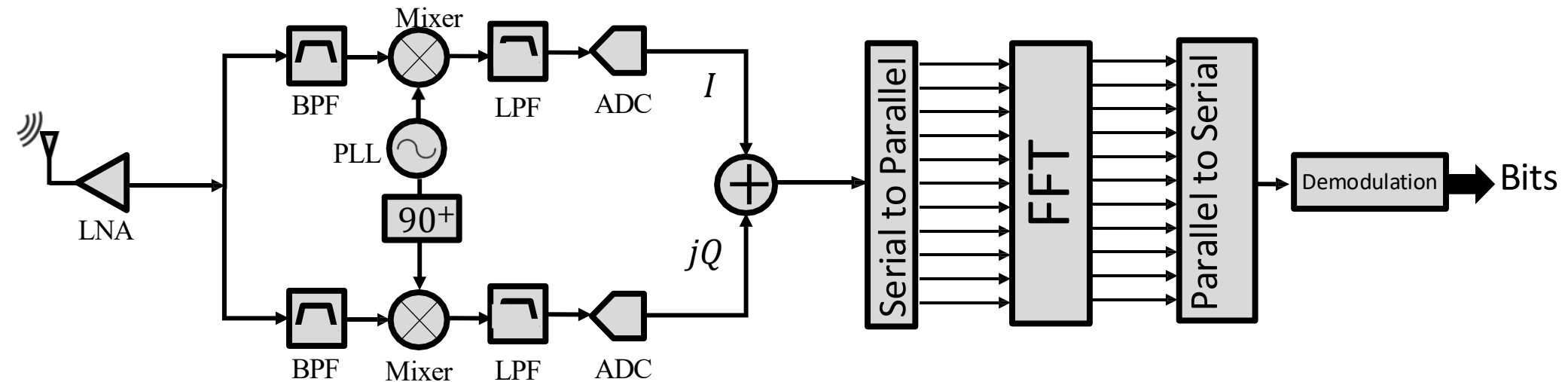
- $N_{\text{subcarrier}} \rightarrow$ IDFT of length N
- Symbols $s[n]$ can come from any modulation: BPSK, QPSK, QAM...
- $x(t)$ is complex $\rightarrow$ need I & $Q \rightarrow$ No point using PAM or ASK ...
- OFDM Symbol: N samples of $x(t)$ generated from the same modulated symbols using IDFT.
- OFDM Symbol Time: $T = N/B$ where B is the bandwidth.
- OFDM Frequency Bin Width: $\Delta f = 1/T = B/N$

OFDM: Orthogonal Frequency Division Multiplexing

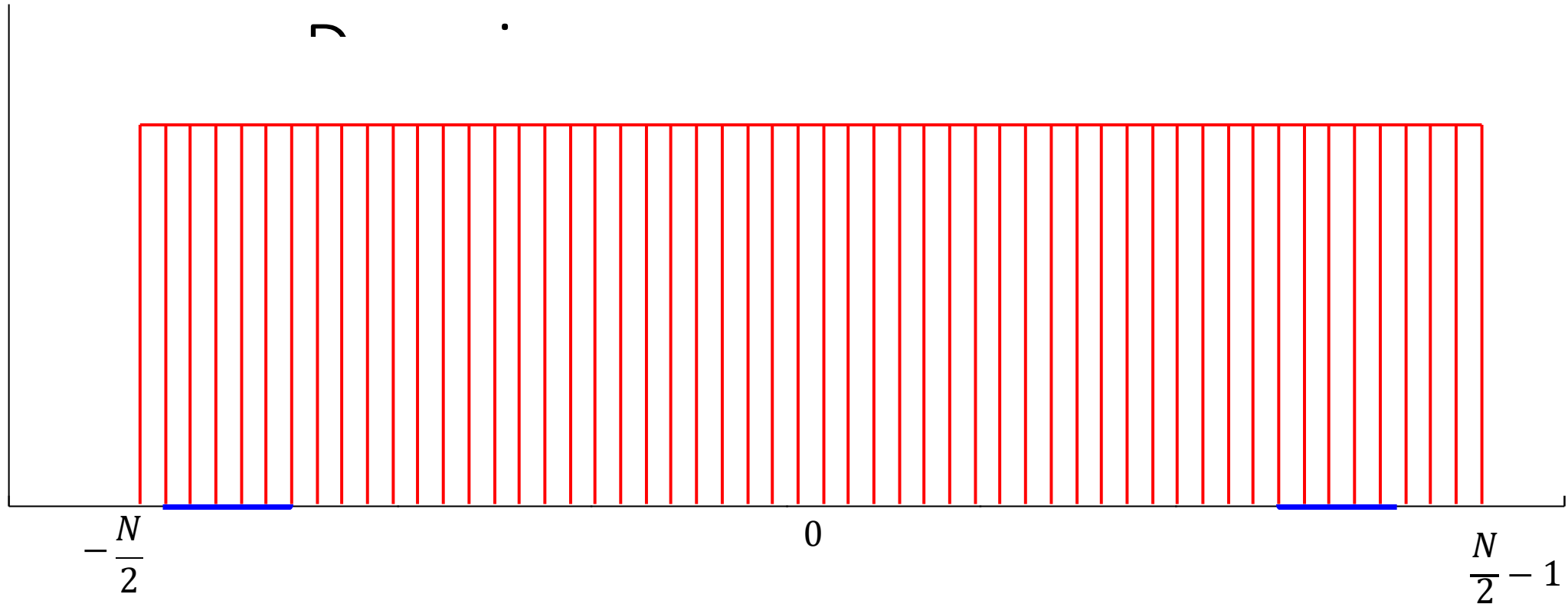
Transmitter



Receiver



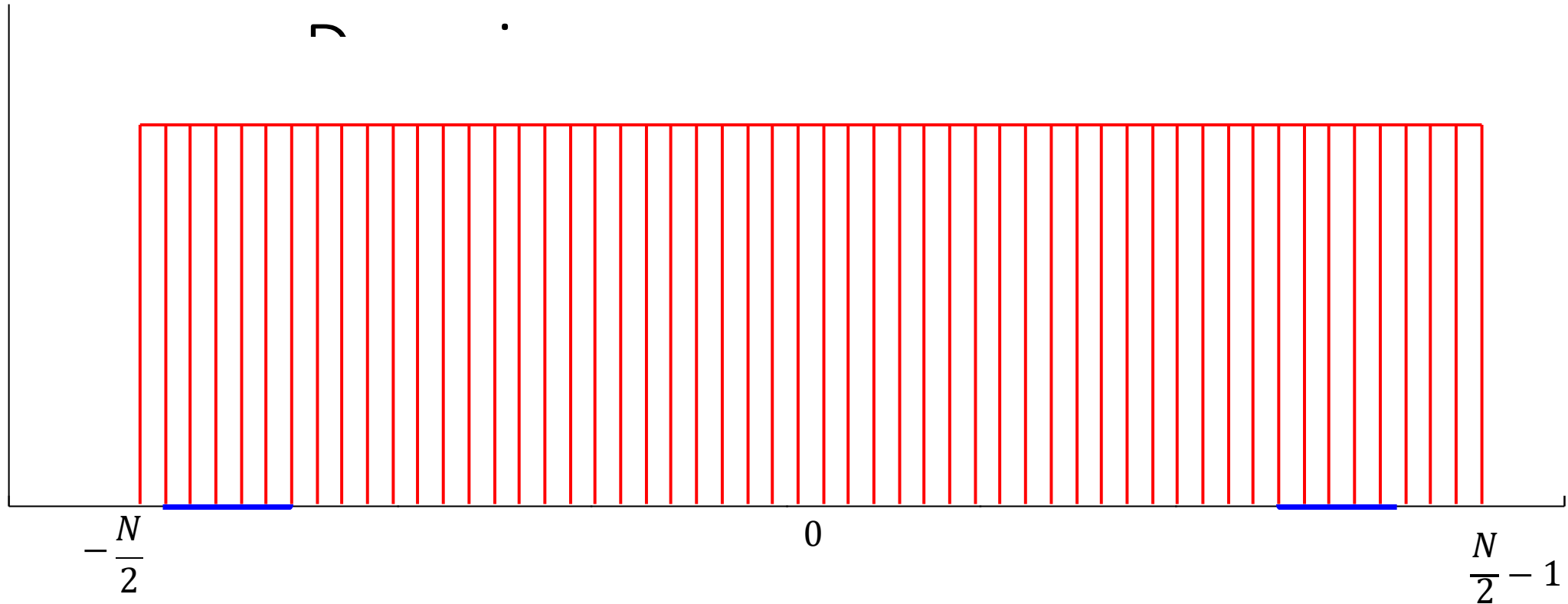
OFDM Symbol in Frequency



- FFT can be represented 0 to $N - 1$ or $-N/2$ to $N/2 - 1$.
- OFDM Symbol created in digital baseband $\rightarrow$ 0 bin corresponds to DC

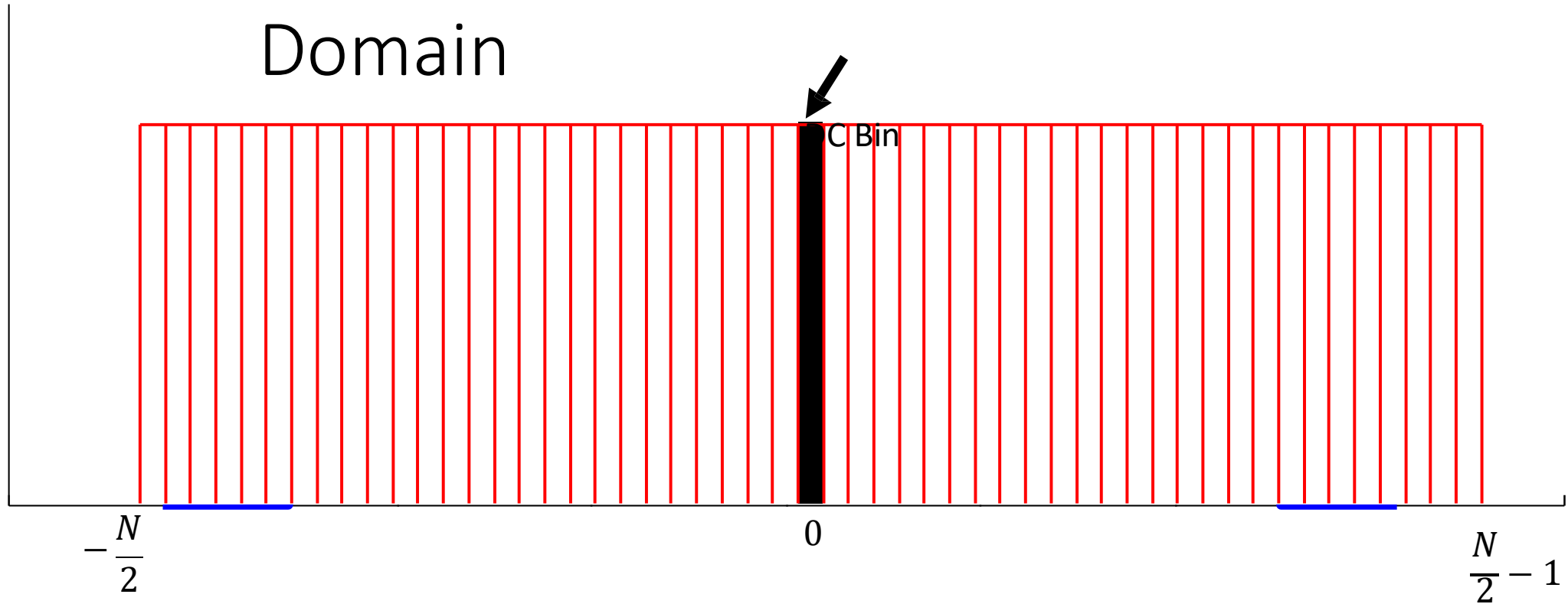
$$X(0) = \frac{1}{N} \sum_{t=0}^{N-1} x(t) e^{-j \frac{2\pi 0 t}{N}} = \frac{1}{N} \sum_{t=0}^{N-1} x(t) = DC$$

OFDM Symbol in Frequency



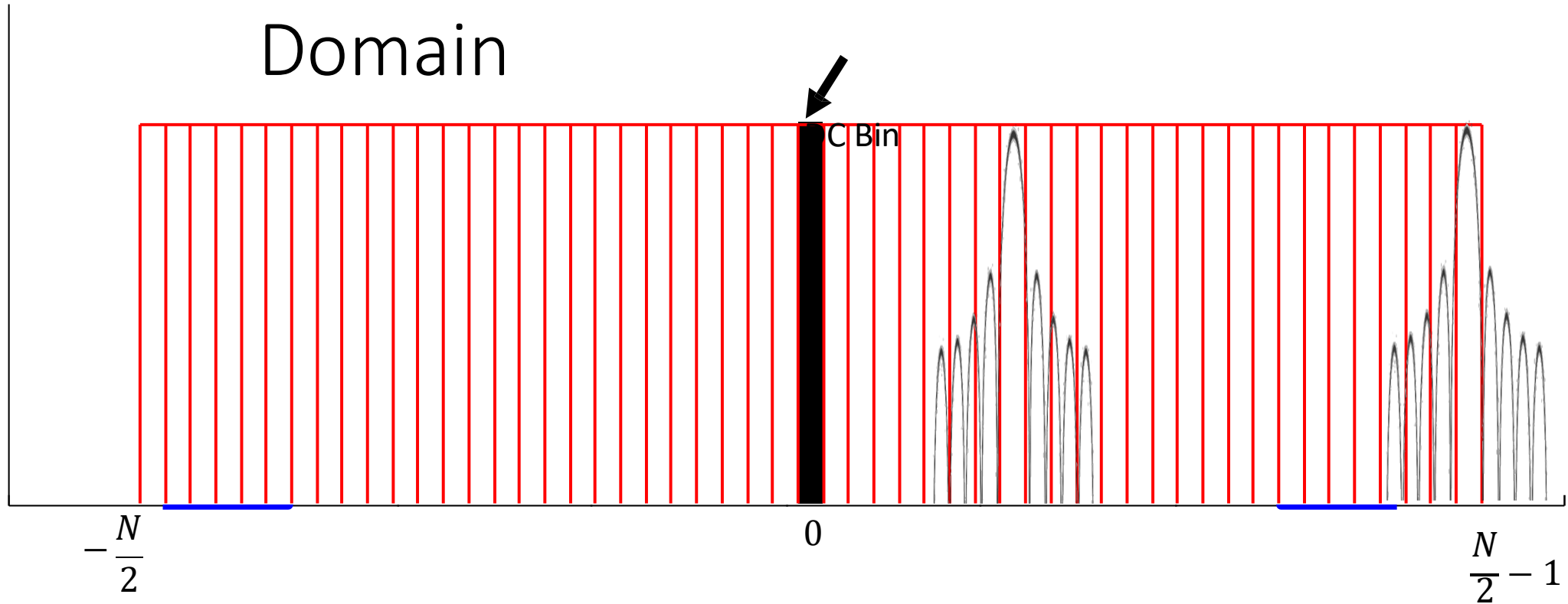
- FFT can be represented 0 to $N - 1$ or $-N/2$ to $N/2 - 1$.
- OFDM Symbol created in digital baseband → 0 bin corresponds to DC
- DC of the circuits corrupts bits sent on the 0 bin → Do not use 0 bin

OFDM Symbol in Frequency Domain



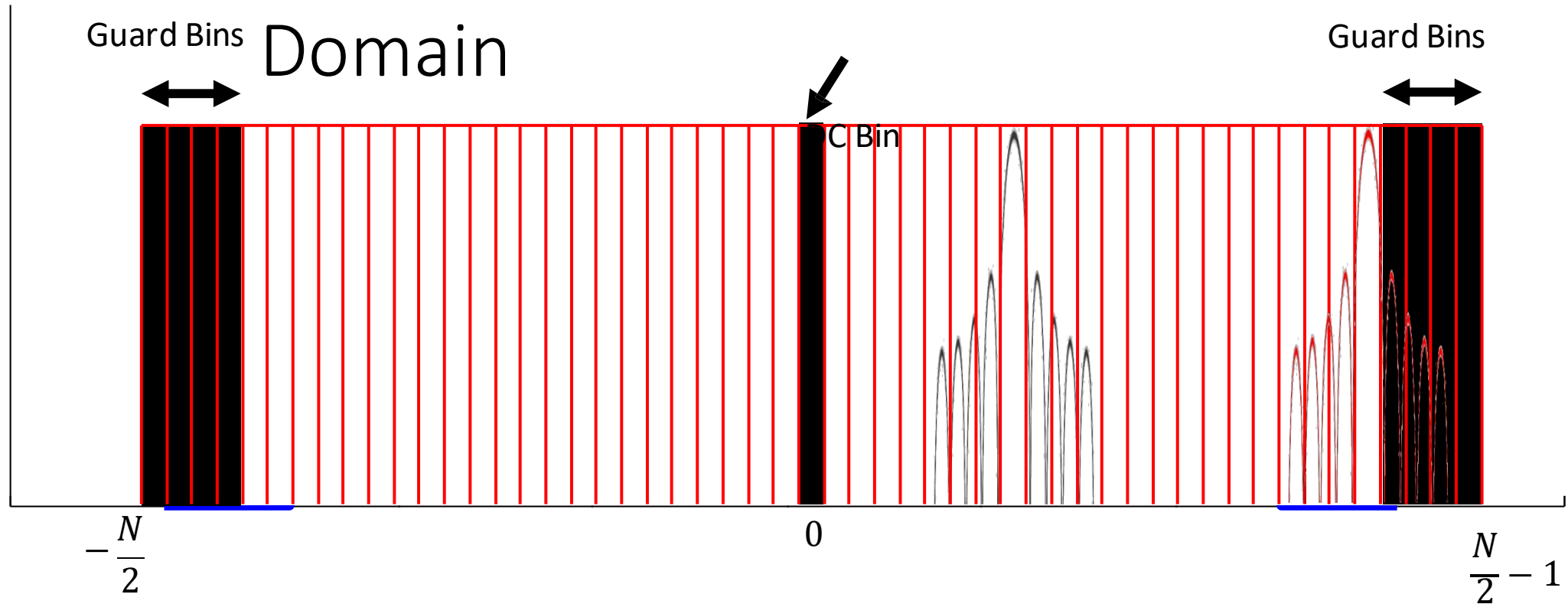
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OFDM Symbol in Frequency Domain



- Subcarriers orthogonal to each other but not to near by channels.
- Need Guard Bins at sides of the channel → Transmit nothing there

OFDM Symbol in Frequency



- Subcarriers orthogonal to each other but not to near by channels.
- Need Guard Bins at sides of the channel → Transmit nothing there
- Reduce Number of Guard band from N to 2 → Very Spectrally Efficient

OFDM: Orthogonal Frequency Division

Multiplexing Transmit Symbols in Frequency Domain On Orthogonal Subcarriers

